Mechanical Engineering Senior Design Project

(ME SDP)

ME SDP128: Formula Racecar Undertray

Design Report

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Executive Summary

Binghamton Motorsports' current Formula SAE vehicle, EV2, is currently in development and is due for an updated undertray design. The undertray is the largest component of the vehicle's aerodynamics package and is responsible for managing airflow under the vehicle. A well-designed undertray can increase downforce while reducing drag.

The previous undertray has been in use since the 2022-2023 season and has experienced years of wear and tear. While the rest of the vehicle has been continuously developed, the undertray has not been studied to investigate potential improvements. The previous undertray also had various shortcomings, including a difficult installation process, failing to utilize the full build zone, a vital region that was removed for jacking-bar access, and non-compliance with FSAE rules. Additionally, CFD results indicate the previous undertray produced -12 Newtons of lift and 26.4 Newtons of drag, for an overall lift-drag ratio of -.45.

SDP128 was tasked with redesigning the undertray for EV2, with the requirements outlining an undertray that is rule-compliant, has an improved lift-drag ratio, and a simplified installation process. Additional stretch goals include a multi-piece design for simplified installation and a lighter undertray, which aligns with the aerodynamics team's goal of a lighter overall package.

The team began by studying the previous design's shortcomings as well as undertrays in motorsports, ground and venturi effect, and the applicable rules outlined by the FSAE for the 2025-2026 season. Once the frame design was completed, an aerodynamic zone was established according to the rules, and the design phase began.

The current design features an enlarged inlet, center constriction, full-length strakes, and a large, gently sloping diffuser. The inlet is designed to capture air flowing below the vehicle's front wing and nosecone. The center section compresses in both width and height, aiming to increase air velocity while decreasing air pressure, including strakes to keep airflow separated and clean. The diffuser aims to gradually increase the pressure before it exits to prevent flow separation. CFD results from the current design indicate the undertray generates -88 Newtons of lift with only 26.6 Newtons of drag, for an overall lift-drag ratio of -3.31.

3D modeling was done using Autodesk's Fusion 360, the software Binghamton Motorsports uses, allowing for seamless cross-referencing with other subsystems. Computational fluid dynamics (CFD) using ANSYS SpaceClaim and Fluent has been employed to study and compare designs to the old undertray, as well as the black box of the car for baseline numbers.

Manufacturing with composites, specifically carbon fiber, requires a mold to be created for the base of the layup. Once the design is finalized, a mold will be created using ½", 1", and 2" foam sections to capture the undertray's geometry. The mold will then be sanded smooth, and the carbon-epoxy layup will begin. The layup will be vacuum-cured, the composite separated from the mold, and sanded to ensure a smooth surface, minimizing boundary layer seperation.

As the undertray will be used by Binghamton Motorsports, the design and manufacturing process follows their schedule, with manufacturing starting in the Spring semester. This is to allow ample testing and validation time. As a precaution, the team has not used any of the available budget, as the design has not yet been finalized; however, material estimates show the budget is sufficient for the entire manufacturing process.

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1. Problem Statement

1.1 Problem Scope

SDP128 aims to develop a new undertray for Binghamton Motorsports' new Formula SAE vehicle, EV2. The previous undertray has been in use since the 2022-2023 season and has been used on two previous vehicles at competition, as well as a third vehicle in testing. In that time, the undertray suffered road rash, improper handling, and was modified on the fly to fit the various chassis. In addition to the wear and tear, the rest of the vehicle, including the aerodynamics package, has been redesigned to meet the evolving goals of the vehicle, leaving undertray improvements unstudied.

Working alongside the aerodynamics subteam, goals were established for the undertray redesign to align with the EV2 aero package goals. The aero team's primary goal is to find ways to reduce the overall weight of the aero package without sacrificing downforce figures or creating excessive drag. The new undertray design targets an improved lift-drag ratio, a simplified installation process, and indisputable rule compliance. Additional feedback from the aero team addressed shortcomings of the previous undertray and requested that those areas be a primary focus of the design team.

As this component will be used by Binghamton Motorsports at the FSAE competition in Michigan, it must comply with all rules and regulations outlined by FSAE. The undertray design and manufacturing process must also follow Binghamton Motorsports' timeline to ensure it is able to undergo proper testing and validation before competition.

1.2 Technical Review

Ground effect theory in relation to race cars is concerned with generating downforce via low pressure on the surfaces nearest to the ground"[Zhang]. The aerodynamic phenomenon can be attributed to the Bernoulli principle, where a velocity increase occurs concurrently with a static pressure drop [Eh]. The goal of an undertray is to act as a suction for the car. This suction is produced by the fast air passing through the undertray, creating a low-pressure region, allowing for a low-pressure region to form (suction). As per Bernoulli's principle *Equation 1,* the faster a fluid moves, the lower the static pressure.

$$P + \frac{1}{2}\rho V^2 + \rho g h = \text{constant} \qquad \textit{Equation 1}$$

A venturi tunnel is a design typically incorporated in undertrays to take advantage of Bernoulli's principle. While air is not an incompressible fluid, and can not be assumed to have a constant density, we can use a constant mass flow rate, simplifying *Equation 1* into *Equation 2,* to explain how narrowing an airflow passage will increase the air's velocity. A venturi tunnel manipulates airflow to increase the fluid's velocity.

$$\rho_1 A_1 V_1 = \rho_2 A_2 V_2 \qquad \textit{Equation 2}$$

A common phrase within motorsports is “Diffusers make undertrays work”. Without a proper diffuser, the undertray will not produce the anticipated results. With respect to *Equation 2*, having a wide inlet/outlet area enables fast-flowing fluid to be manipulated and behave more predictably. Not only does the diffuser have to allow for a large mass flow rate, but it must be angled properly to prevent downforce losses due to flow separation and vortices forming. Diffuser angles have been heavily researched; while the exact angle depends on the diffuser length and accompanying vehicle/undertray, angles should not lie outside of 5°-20°[Eh]. Due to how dependent the optimal angle is on other components, an iterative development using simulations is the best approach to finding that angle.

According to the 2025 FSAE regulations, the undertray must remain within the designated boundaries, contain no moving components, include drain holes, allow for timely measurement of ground clearance, and be electrically grounded.

In reference to Binghamton FSAE’s last undertray design, our goal is to better implement the existing designs using the knowledge gained from research and iterative design using CFD. We plan on using better manufacturing and sanding processes for the new undertray in order to prevent the losses due to skin friction that the old design had. Additionally, the old undertray’s grounding mechanism was ineffective, unprofessional, not very presentable, and could be improved upon using a more concise and effective design. Furthermore, the old undertray did not implement drainage holes or an area designated for judges to have ease of access to measure the ride height, which has been an issue plaguing the previous undertray's design.

2. Design Requirements

The FSAE undertray design is primarily bound by eight major requirements, driving the final design to have significant improvements across all relevant fields.

The first requirement, *[MESDP-FRU-01],* requires the undertray to be composed of multiple pieces. A major issue with the previous design is that it required several people to install it on the frame of the car, and the process was extremely lengthy. Keeping the new undertray in multiple pieces, fewer than four, allows for easy installation with minimal personnel.

The second requirement, *[MESDP-FRU-02],* states that the undertray should not interfere with other subteams’ components. An undertray is typically the last component of the car’s design. This design should not require other subteams to make adjustments to accommodate the undertray; rather, the undertray will be changed in order to satisfy the needs of components with greater importance. Specifically in this new model, part of the motor protrudes below the frame; the undertray should be adjusted around this, rather than forcing an impractical change in the motor design.

The third requirement, *[MESDP-FRU-03],* requires the undertray to be grounded. The previous model did not originally include this feature, which led to a last-minute adjustment, ultimately compromising some of the effectiveness of the design. Ideally, the

undertray will be grounded via hardware that's already being used, as long as this doesn't interfere with ease of installation. This is also an FSAE requirement contributing to the eligibility of the car at competition.

The fourth requirement, *[MESDP-FRU-04],* is that the new undertray shall have a lift/drag ratio improvement over the previous design. The ideal target is a ratio of two or greater, although improvement above the original design is a non-negotiable requirement of the new design.

The fifth requirement, *[MESDP-FRU-05],* states that the new design should weigh less than the previous undertray. This will allow for the improved dynamic performance of the car, as well as improve downforce effectiveness. Both should be weighed upon completion.

The sixth requirement, *[MESDP-FRU-06],* requires that the undertray shall be removed from the car using only basic hand tools. This will allow for ease of installation and removal, which is vital when doing maintenance. Additionally, using simple hardware allows for an installation and removal process that requires little to no training.

The seventh requirement, *[MESDP-FRU-07],* states that the undertray shall adhere to all applicable rules outlined in the FSAE 2025/26 Rulebook. If the undertray does not meet FSAE rules, it will not pass inspection, which disqualifies the car from competition. Undertray rules are listed explicitly in the FSAE rulebook, quantifying objectives like size restrictions, leading edge sharpness, and distance from the ground.

Lastly, the eighth requirement, *[MESDP-FRU-08],* states that the undertray shall not exceed the allocated budget. The SDP128 project has been given a $2,000 budget. Any and all expenses must be approved and remain below this final budget number. In the case of going over budget, the undertray would be incomplete and unable to be used.

3. Design Description

3.1 Overview

This section will cover the overall design components and summarize their importance, covering the overall undertray, venturi tunnels, and rear diffuser.

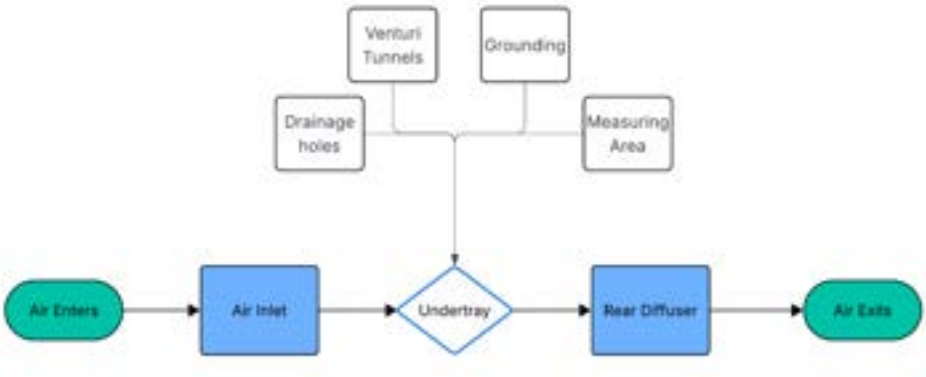


Figure 1: Design Flowchart

The new undertray's goal regarding airflow is to have the air pass from under the nosecone, through the flat floor, through venturi tunnels to pick up speed, and exit through the rear diffuser with minimal to no flow separation or flow vortices.

The speed increase throughout the undertray creates a low-pressure region, ultimately sucking the body to the ground, and the rear diffuser allows for smooth pressure recovery, dispersing exiting air without flow separation, which allows the undertray to maintain a low-pressure system.

Sealing strategies help ground effect aerodynamics by maintaining pressure regions, which maximizes downforce. Skirts help seal air off by creating an extended surface, preventing leakage. Vortex sealing is another approach that seals off the low-pressure region by using complex geometries to generate vortices, creating dynamic barriers. Side ramps are a type of vortex sealing that prevents lateral airflow leakage while blocking tire wakes from interfering with the air below the undertray.

The downforce concentration is expected to be at the center of the undertray, where the fastest-moving air is, hence the greatest suction. As the air passes through the rear diffuser, the air should gradually increase in pressure for a smooth pressure recovery.

An important variable that must be considered when designing an undertray is the tire wakes. The air passing over and around the tires of the vehicle is turbulent and creates vortices, while the undertray depends on smooth airflow. Undertrays can be designed to direct this turbulent air away from critical regions to avoid disturbance to downforce.

In order to be FSAE rule-compliant, the undertray should include drainage holes, be electrically grounded, and allow for an examiner to measure the ground clearance with ease. To meet these requirements, the top of the undertray is designed with a gradient to allow leaking fluids from the car to find the drainage holes, which double as measurement holes. The undertray is also designed to be electrically grounded. The design includes the implementation of a wire connected to the frame and a split-bolt, containing a wire, connected to the frame, which is held in a fixed position.

3.2 Detailed Description

3.2.1 Geometry + Airflow Path

A flat undertray creates a stable and reliable foundation for the undertray. If the undertray has a curved bottom, there will be even more unpredictable airflow, which can put the driver and car at risk. With a curved bottom, the airflow is at risk of losing laminar flow to turbulence and vortices due to its uneven nature. Furthermore, a curved undertray would require more material, which adds more weight. A major goal of the aerodynamics team is to decrease the overall weight of the aerodynamics package, and with the undertray being the largest aerodynamic component, it is essential for it to weigh less than the previous design.

The inlet is designed to be flush with the nosecone, as the nosecone is located at the front of the vehicle, and most of the air the inlet receives would have already been redirected by the nosecone.

The Venturi tunnel is a widely used design, configured to take advantage of Bernoulli’s principle. According to Bernoulli's principle, if a fluid speeds up, the gain of dynamic pressure must be compensated by a loss of hydrostatic pressure (elevation-related) or by static pressure. Given there is no change in elevation, the static pressure must drop. With a low-pressure region below the car, the undertray is sucked towards the ground, generating the aspired downforce.

The rear diffuser is set at an angle of 10° to allow for the best possible pressure recovery. Through iterative design and CFD simulations, 10° has so far been our best angle, preventing flow separation and vortices from forming.

Side ramps may still be implemented to prevent any losses that may result from tire wake and lateral leakage. The design takes the air from the tire wakes and directs it towards a ramp that acts as a downforce component. They would be angled so that the turbulent air is utilized to add downforce rather than interfere with the laminar flow beneath the vehicle. Additionally, the side ramps would extend laterally and create a wall for any air that tries to escape to the regions adjacent to the side of the car.

Throat height varies between 40- 50 mm, which is limited by FSAE maximum ride height rules (90 mm) and the recommended 75 mm. At this height, all the other ground effect applications will be able to function properly. The throat height is not as effective in

creating downforce, but is intended to be low enough for the undertray to create a suction, yet not be too low as to choke the undertray from receiving any air.

3.2.2 Structural Considerations

The undertray was manufactured using a carbon-epoxy composite, where carbon fiber is combined with epoxy resin, forming a rigid solid. Carbon fiber is a material made of very thin and strong filaments of carbon atoms, bonded to each other in long chains that are typically about 5-10 micrometers in diameter [2]. As indicated in *Figure 2*, this bond makes the material roughly five times stronger than steel, twice as stiff, with a notably high tensile strength [3]. Compared to aluminum, it offers significantly more strength and stiffness while being about 40% lighter [4]. For this reason, in applications where minimizing weight and maximizing performance are vital, carbon fiber can be found in various high-performance vehicles like racecars, aircraft, and bikes.

Carbon fiber and other materials properties

Material	Tensile strength (MPa)	Tensile modulus (GPa)	Elongation (%)	Density (g/cm³)	Strength-to-weight ratio (MPa/g/cm³)
Carbon fiber (1)	5407	294	1.75	1.79	3026
Kevlar 49 (5)	3650	124	2.5	1.44	2535
S-Glass (5)	4600	88	5.5	2.49	1847
Boron (5)	3620	400	1	2.57	1409
Titanium alloy Ti-6Al-4V (type 5) (2)	620	113.8	14	4.43	140
High-Carbon Steel (4)	1010	198	14.6	7.49	135
Aluminum alloy 6061-T6 (3)	310	68.9	17	2.7	115

Figure 2: Carbon Fiber and Other Material Properties [3]

To form a more rigid composite material, the carbon fiber can be saturated with epoxy resin, which is a type of thermosetting polymer resin. The epoxy in these composites acts as a glue that binds the carbon fibers together and distributes the stresses between the fibers in a more uniform manner. The resin is highly sought after in these applications due to its strong adhesion to carbon fiber, high mechanical strength, rigidity, favorable chemical and environmental resistance, and low shrinkage upon cure [5]. When used in a carbon-epoxy composite, the epoxy can help produce an end product that is lightweight, stiff, and durable.

The composite layup process begins by layering sheets of dry carbon fiber. This process, also known as a wet layup, is a less expensive option that allows for minor adjustments to

the fiber sheets prior to the epoxy application. Additionally, by layering the carbon fibers in a unidirectional (0°) manner, which means the fibers are aligned and oriented in the same direction, the stiffness and strength can be further maximized.

Once completed, the undertray is then attached to the chassis at eight main points, with four on each side, for a more secure and stable mounting when distributing loads. These are attached through nuts and bolts to reinforce the tabs that attach to the car's frame, and flexible cables or straps may also be utilized to help with both the movement and installation process. Additionally, by evenly distributing the mounting points across the undertray, any stress concentrations in an area can be reduced, which in the end also prevents the possibility of damaging the carbon-epoxy composite [6].

The expected loading conditions for the undertray are still being refined with ongoing CFD simulations. In developing the preliminary design, it must be considered that carbon fiber composites perform best when loads are distributed, as they are less tolerant of concentrated point loads. Once the CFD results are finalized and the aerodynamic loads are fully defined, a finite element analysis can be performed to evaluate the structural response of the undertray on a more realistic level.

3.2.3 CFD Development

In collaboration with two members of the aerodynamics team, CFD analysis was performed on ANSYS Fluent to test and refine the existing undertray designs. The process of CFD can simulate how airflow would act through the undertray, which will help determine the performance factors of downforce, drag, and cooling. This allows any performance issues to be highlighted without having to make multiple physical prototypes of the undertray, since that would be both time-consuming and not cost-effective.

Once the CAD models were completed on Fusion, they were simplified to reduce sharp or overly complex geometry. They were then imported into ANSYS SpaceClaim for cleanup and region definition. The computational domain was set up, with mesh sizes ranging from 1 mm around the undertray to 16 mm in the outer enclosures, with four boundary conditions applied for accurate wall flow resolution. Boundary conditions were then set with a 20 m/s velocity inlet, a 0 Pa pressure outlet, and the aerodynamic surfaces. The simulation was run with convergence checks, and initial results included surface pressure coefficient distributions and pathline visualizations. These showed the low-pressure areas responsible for downforce and highlighted regions where flow separation may occur. Further results are still being generated, and the full analysis is currently ongoing.

3.2.4 Packaging Constraints

While the undertray is mounted to the bottom of the chassis, it can still interfere with other components on the chassis. The primary constraint impacting the design of the undertray is the FSAE rules that outline the zone the undertray may occupy. This zone is determined by both the frame dimensions and the wheel locations. The aerodynamic zone is defined by the following description: [1] "No more than 700 mm forward of the fronts

of the front tires, no more than 250 mm rearward of the rear of the rear tires. No part of the vehicle may enter a keep-out zone defined by two lines extending vertically from positions 75 mm in front of and 75 mm aft of the outer diameter of the front and rear tires in the side view elevation of the vehicle, with tires steered straight ahead. This keep-out zone will extend laterally from the outside plane of the wheel/tire to the inboard plane of the wheel/tire." The following image, *Figure 3,* is supplied by the FSAE rules and provides a visual representation of the aerodynamic zone.

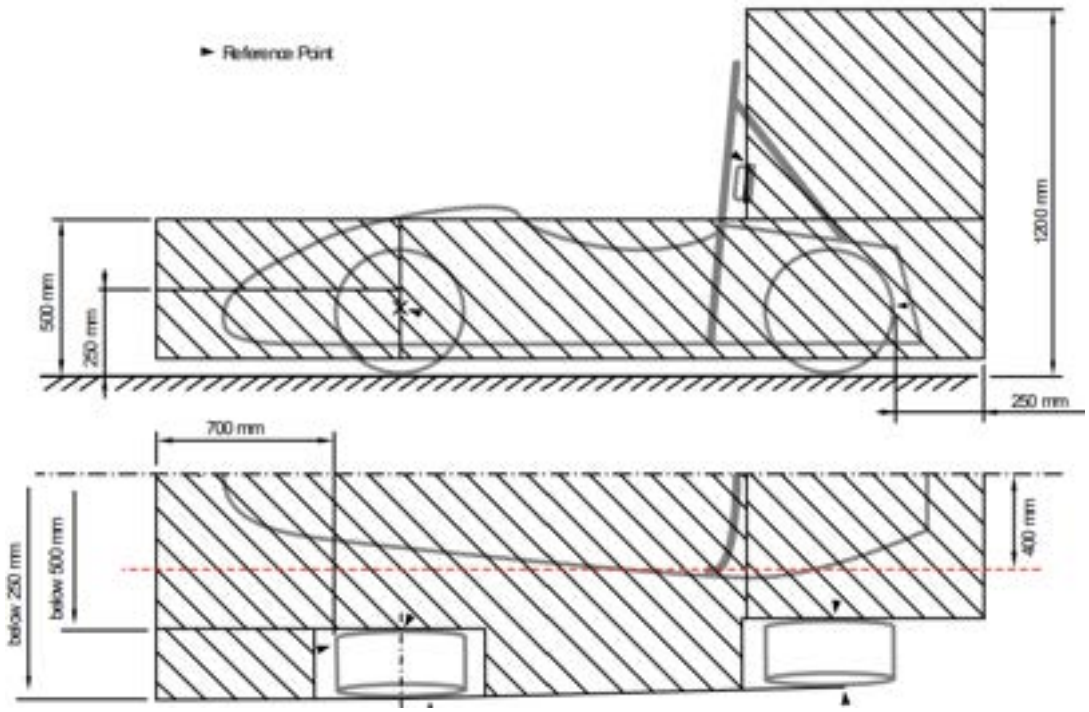


Figure 3: Aerodynamic zone visual aid as supplied by Formula SAE Rules 2026 [1]

With the aerodynamic zone defined, several chassis and suspension components were identified within this region, necessitating the undertray geometry to be designed to accommodate them. The front suspension, specifically the lower control arms, limited the undertray's forward length. The lower control arms are dynamic components that rotate with the wheels as the vehicle's suspension is compressed (during acceleration, braking, and cornering), and sufficient clearance is needed to prevent the components from contacting the undertray. In addition, the front tires need to be able to turn from full left to full right without interfering with the undertray, adding to the limit of how far forward the undertray could extend, as can be seen in *Figures 4 and 5*.



Figure 4: Isometric view of front-lower control arms in relation to the undertray.



Figure 5: Top view of front-lower control arms and wheels in relation to the undertray.

While the front suspension limited how far forward the undertray could extend, the nosecone helps to direct air to the undertray, reducing the losses due to the front suspension.

Moving rearwards, the next obstacle that impacted the usable zone for the undertray's design is the driver's seating area of the frame. From the side view, the cockpit area is the lowest point on the frame, which forced the undertrays' center-section height to be reduced. This aided in the venturi tunnel effect of the design; however, it introduced ground-clearance concerns during the design process. The following image, *Figure 6,* shows the side profile of the EV2 frame.



Figure 6: Side profile of the Binghamton Motorsports EV2 Frame.

Continuing rearwards, the rear suspension and frame rails limited the maximum angle achievable by the diffuser. As the rear lower control arms are also dynamic parts, clearance is needed to prevent contact with the undertray. The rear lower control arms can be seen in the following images, *Figures 7 and 8.*

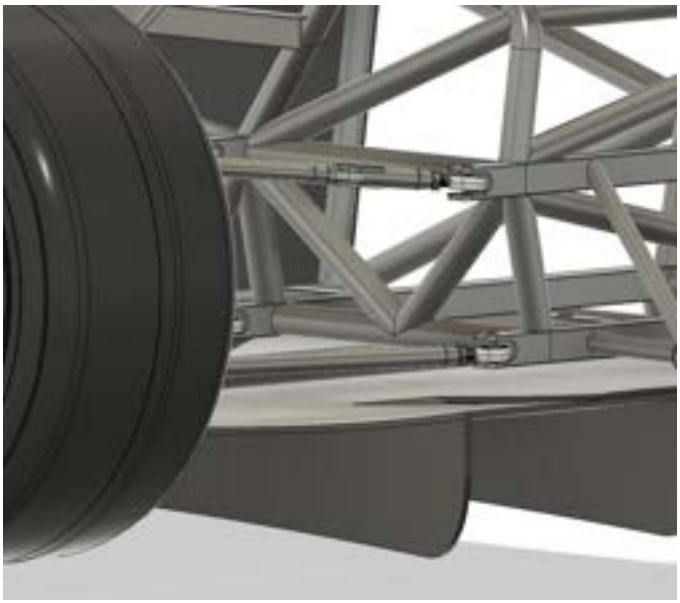


Figure 7: Rear isometric view of the rear upper and lower control arms in relation to the undertray.



Figure 8: Side view of the rear upper and lower control arms in relation to the undertray.

Inward from the rear suspension lies another design hurdle, the electric motor, visible in *Figure 8.* The motor sits lower than the frame rails it is mounted to, and acts as the real limit of the maximum angle the diffuser could achieve. The workaround was to put a small bulge in the design, just large enough to prevent the motor from contacting the undertray, which can be seen in *Figure 9*.



Figure 9: Rear view of the undertray, showcasing the bulge for the electric motor.

The final packaging constraint is the rear jack-bar, visible in figures 10 and 11.

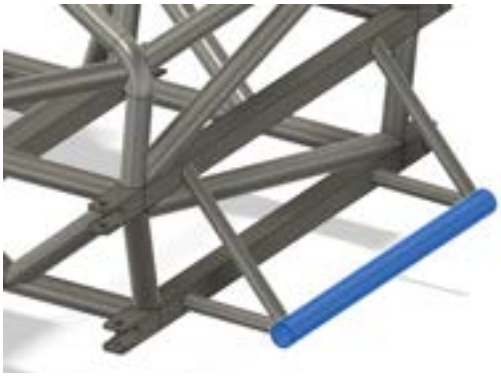


Figure 10: Isometric view of the rear jack-bar (highlighted).



Figure 11: Side view of the rear jack-bar.

The jack-bar needs to be easily accessible, which, in previous years, meant the diffuser was cut off to allow for room around the bar, as can be seen in *Figure 12*. The EV2 undertray design uses a separate, removable piece below the bar to allow the diffuser to span the entire width of the undertray, which can be seen in *Figure 13*.

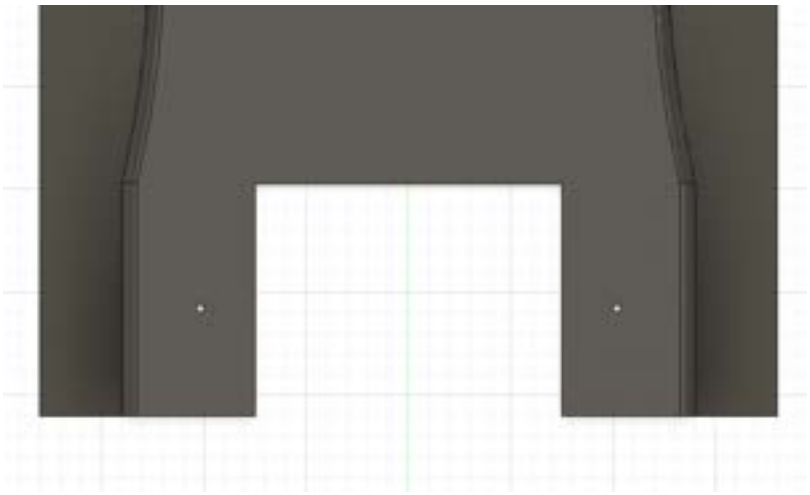


Figure 12: Top view of the previous undertray design, which sacrificed the diffuser area for the rear jack-bar.

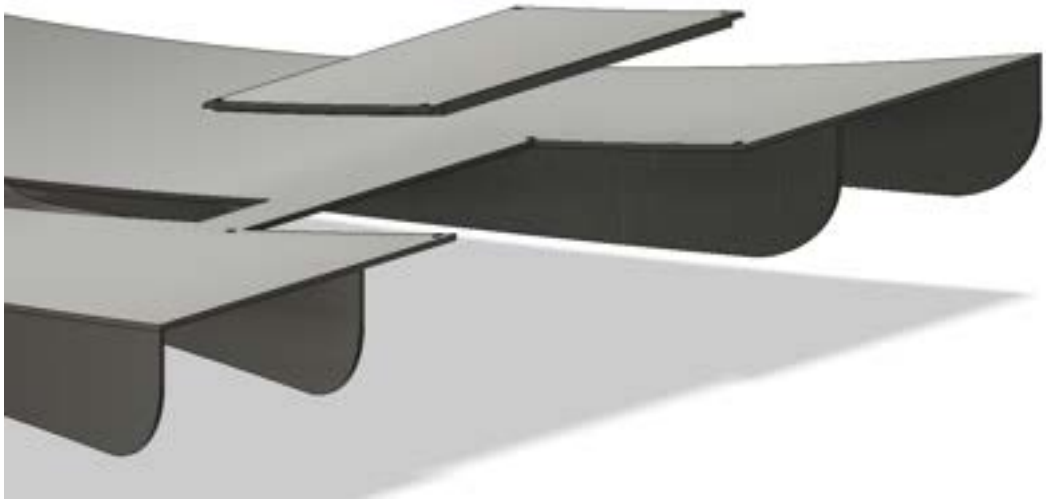


Figure 13: EV2 Undertray design, with removable diffuser piece.

3.2.5 Manufacturability

The undertray will be manufactured using a negative mold to ensure the aerodynamic surfaces are accurately represented. The mold will be constructed from assembled sections of foam insulation board ranging from 0.5-inch to 2-inch sections as necessary. The process begins by creating .DXF files of the cross-sectional profile of the undertray, at every 0.5-2 inches along the length, see *Figure 14*.



Figure 14: Highlighted cross-section of undertray to be exported as a DXF

These .DXF files will be verified for errors and sent to Binghamton University's waterjet machine to be cut into the desired profiles. Each board will be labeled and contain locating holes to line up each piece accurately and in order. The individual profiles will be assembled with wood dowels through the locating holes and glued between each foam layer, as shown in *Figure 15*.



Figure 15: EV1 Undertray Complete Negative Mold

For complex geometric areas, .stl files are created for 3D printing, then attached to the complete foam structure to ensure greater accuracy. The final mold surface is prepared through a lengthy sanding process, progressing through finish grits ranging from 180 to 800+. Any assembly gaps or errors are rectified using spackle before the final smoothing steps. The overall dimensions of the mold are roughly 1763.00 mm x 1125.80 mm x 97.724 mm, which have been confirmed to fit within the constraints of the waterjet when cut cross-sectionally.

Furthermore, the carbon undertray will be fabricated using a wet carbon layup process and finished through vacuum bagging. Before any composite is laid, the mold is prepped with a PVA release agent. Carbon fiber sheets are cut and test-fitted to determine areas that require more focus during the application. The chosen epoxy/hardener mixture uses a slow-curing hardener to maximize the team's working time during the application. This will allow for between 45 and 60 minutes to complete the layup, and likely more time if epoxy is mixed throughout the process rather than all at once. The layup sequence consists of three structural carbon layers, each followed by resin: Layer A of carbon and epoxy, Layer B of carbon and epoxy, and Layer C of carbon and epoxy. This part of the process is tedious and will require multiple people to complete, as seen in *Figure 16*. This structural layup is then covered with a peel ply sheet and breather cloth before being sealed in a vacuum bag and subjected to vacuum pressure. After curing, the final product will be trimmed, wet sanded, and cut into the desired pieces for ease of installation.



Figure 16: EV1 Undertray Carbon Layup Process

The primary risk with manufacturing is ensuring the accuracy of the mold surface, particularly in creating accurate .DXF files, and assembling each foam profile in order and in line. Inaccuracies in these areas could lead to inconsistent airflow and affect the aerodynamic performance, causing digital simulation results to be inaccurate. The vacuum bagging process mitigates the risk of air pockets, which is particularly important to the final finish. With proper setup, air pockets will be minimal, and final sanding will be a smooth process.

3.3 Use

The undertray is installed beneath the chassis and acts as a passive aerodynamic component that shapes and controls the airflow under the car to enhance downforce and prevent any air from entering the engine bay or cockpit, which would generally increase drag. The design itself is fully static, requiring no user adjustment when used, and interfaces automatically with the airflow whenever the car is in motion. Additionally, for ease of access and maintenance, the undertray design is divided into two sections and engineered for quick and easy removal using basic hand tools. Practical ground clearance considerations have also been integrated, with removable rear sections to allow better clearance over speed bumps and ramps. The intended ride height is also around 3 inches to the lowest frame point, which ensures safe operations on both the track and paddock area.

4. Build and Implementation

The manufacturing process began by creating .DXF files of the undertray geometry to create the mold. The first step in creating the .DXF files were used to determine the optimal thickness each section should be, keeping in mind both the overall geometry and

the cost of the material to be cut. Given the geometry, it was determined that using 1-inch sections, running the length of the undertray, would best capture the geometry and minimize the sanding needed to prepare the mold for the layup.

The .DXF files were made to size using a *Front View* to capture the height and width of each 1-inch section of the undertray. They also featured a .25-inch square removed from the center of the bottom, visible in *Image 17,* which allowed dowels to be used to align each section. In total, there were 74 1-inch sections created, with 18 additional, 2-inch tall, 5-mm wide sections for the strakes. The Fusion model of the final foam sections can be seen in *Image X.* With all sections created, they were accordingly named 1-74, and laid out on 4’ x 4’ squares to determine the layout in which they would be cut using the Fab Lab’s waterjet.



Figure 17: Complete undertray DXF assembly

The files were then submitted to the Fab Lab’s Manufacturing Request Form and cut on the waterjet. The individual sections were then aligned and glued with foam board adhesive to create larger sections, which were then finally glued together. Once all sections were assembled, the rectangular sections were sanded until they blended together. Any gaps between sections or chunks that were accidentally removed were filled with spackle.

It was at this point that we realized that there was some oversight regarding the 5 mm foam sections for the strakes. While they were cut perfectly, there was not enough rigidity for them to hold their shape, nonetheless survive a layup or being pulled under vacuum. It was decided that to stay on schedule, the main part of the undertray layup would continue, and the strakes would be made separately and attached to the rest of the undertray. Additionally, the inner strakes were swapped out for a singular re-energizer, which can be seen in *Image X,* which acts to further constrict the airflow and further lower the pressure of air in the center of the undertray. The re-energizer was modeled on Fusion, exported as a .DXF, and cut into a 1-inch thick section and 2 ¼-inch sections.

Once the undertray mold was blended and spackled, it was sanded from 200-600 grit to ensure a smooth, level surface with no major issues. The mold was covered in a release film, which allows for the composite to be removed from the mold with little effort. Before the layup, each layer of carbon fiber, the peel ply, bleeder cloth, and vacuum bag was cut to size to streamline the layup process. The carbon was also weighed, and notes were taken on the weight of each layer to allow for a 1:1 weight ratio of carbon fiber and epoxy to be used.

The layup began by applying a layer of epoxy to the mold to make the first layer of carbon fiber stick to the mold. The first layer was then applied, followed by more epoxy, which was spread throughout until all of the carbon fiber was saturated. Layer two of the carbon was then laid on the mold, followed by more epoxy, and repeated for the final layer. Once all three layers were laid on the mold and saturated with epoxy, the peel ply sheets were laid on top, followed by the bleeder cloth, and finally the vacuum bag. The vacuum bag was then connected to the vacuum pump, and the vacuum was pulled to -15 psi and left to cure under vacuum overnight.

The following day, the peel ply and bleeder were pulled off, and the composite was removed from the mold. Immediately, the clear excess carbon was trimmed from the composite, and the composite was put back on top of the mold for sanding. Sanding started with 180 grit and was gradually increased up to 600 grit. With the undertray sanded smooth, focus shifted back to the strakes.

Initially, the strakes were 3D printed based directly on the Fusion model of the strakes, but would have been very difficult to lay up and mount to the undertray. The strakes models were then used to create a negative mold, which can be seen in *Figure 18.* In addition to the negative mold, the original 3D print of the strakes will be used as reinforcement for the strakes and will be wrapped in carbon fiber on the negative mold. The mold also features a face that extrudes inward (when on the undertray) and will act as a lip, which will be used to epoxy the strakes to the main part of the undertray.



Image 18: Negative strake mold (grey) and positive strake mold (black)

The strake layup process began by applying wax to the mold to help release the composite once it was cured. To further help release the composite, a peel sheet was later applied to the mold. It was determined that the easiest way to conduct the layup would be

to lay one large sheet of carbon over both sides of the mold first. With epoxy on the mold, the first layer of carbon was laid down, and the positive mold pieces were positioned in the negative mold to press the carbon into the corners, and later add strength to the strakes.

Once the first part of the layup was cured, the second layer of carbon was again draped over the entire mold and epoxied to the 3D print. The excess carbon was epoxied to the side of the mold, which would later act as the flange used to mount the strakes to the undertray. After 24 hours of cure time, the composite was removed from the mold, and the two strakes were separated using a Dremel to cut down the centerline of the part. With each strake now separated, the clear excess was trimmed away from the strake.

It was determined that it would be easier to attach the re-energizer before the strakes to avoid having to reach over them, so the foam piece was epoxied down to the main part of the undertray and left to cure for 24 hours. Once it had cured, the necessary carbon was cut and draped over the foam to ensure it would form to the foam well. Then, as with any other layup, the foam and surrounding composite were covered in epoxy, the carbon was laid over top, and more epoxy was added and squeezed into the carbon. The following image, *Figure 19,* shows the undertray with the re-energizer immediately after applying the epoxy.



Figure 19: Re-energizer immediately after being attached to the undertray

After 24 hours for the epoxy to cure, the strakes were ready to be attached to the undertray. It began by measuring how far back they would start relative to the undertray according to the CAD model of the undertray mold. With the position figured out, the undertray was removed from the mold to allow clamps to be attached to the strake and the top side of the undertray. A small strip of carbon was cut to be used to attach the center of the strake, as the strake mold did not allow for a full-length lip to be made. To attach the strakes to the mold, epoxy would be poured onto the undertray, and the lip

positioned on the epoxy, with clamps spaced accordingly to hold the pieces together while the epoxy cured. The process was then repeated for the other side, and *Figure 20* shows the first strake attached, and the second strake curing with the clamps holding it on the mold.



Figure 20: Strakes midway through the process of attaching them to the undertray,

Once both strakes were cured in position, it was decided that adding a final layer of carbon fiber to the underside of the undertray would eliminate the need to sand the underside and yield a far more visually appealing final product. The surface was cleaned, epoxy was poured out on the entire surface, and the final layer was draped over the surface. The carbon was continued up the inside surface of the strakes as well, and the transition from the undertray to the strake required extra time to ensure a smooth transition. The following image, *Figure 21,* shows the undertray immediately after the final layer was added and had cured.



Figure 21: Undertray with final layer of carbon fiber applied for both aesthetic purposes and reduced surface roughness.

With all layups done, it was time to trim the excess carbon fiber visible in *Figure 21.* A straight edge was used to mark a line between the tail ends of each strake, which was then cut along using the dremel. There was a small amount of excess carbon outboard of the strake, which was also removed using the dremel.

With the excess removed, the undertray was ready to be mounted to the car. Installing the undertray required lifting the front of the vehicle with a jack, sliding the undertray under the vehicle, and then setting the vehicle back on the ground. The undertray position was determined by centering it relative to the frame and setting it back by measuring its distance past the jack bar in the rear. Once its position was determined, the four main mounting holes were drilled, and the undertray bolted to the frame.

Unfortunately, in March, a decision was made by the drivetrain subteam to change the motor's mounting location to the drivers right hand side, opposite of the original location. Due to a lack of communication, this change was never mentioned to SDP128, and it was only noticed when mounting the undertray. A small rectangular area was cut out of the undertray to clear the motor. In the future, the original hump intended to cover the motor can be cut out and riveted to the correct position, and a flat panel used to cover the hole left on the drivers left hand side.

With the undertray securely mounted to the frame, the front corners of the undertray would have to be trimmed down to allow the front tires to turn from full left to full right.

To mark the route traveled by the tires, the front undertray bolts were removed, which allowed the undertray to sag in the front, and the car jacked up to bring the tires above the undertray. Holding a sharpie tangent to tire and perpendicular to the ground, the wheels were turned and the routes traced on both sides. To increase clearance, another line was traced 1 inch further inboard of the path the tires would travel. Using a Dremel, the undertray was cut along the line further inboard, and both front corners were removed. The undertray was reattached, the car lowered, and the wheels turned with the undertray in place to confirm the wheels cleared the undertray.

The corner mounts were the next step, which prevent the undertray from deflecting down and contacting thr ground, but allow it to deflect up in the event it does contact the ground. These mounts use eye bolts, safety wire, and chain connectors to allow for quick removal of the corner mounts. The position of the eye bolt needed to be as close to possible as the corner to provide the necessary support, so holes for the bolts were drilled 1 inch inboard to the centerline of the frame, and 1 inch inboard along the length of the frame. Using the safety wire and cable clamps, a loop was created on one end of the wire, and that end attached to a nut and bolt, with the front being attached to the node for the coilover assembly, and the rear attaching to the rear wing-rod bolts. The wire was then measured to the head of the eye bolt and cut. Chain connecters were then attached to the end of the wire using another cable clamp. Finally, the chain connecter was attached to the eye bolt. One all four corners were set up, the length of the wires were adjusted to hold the undertray up and leave 1 inch of ground clearance between the bottom of the strake and the ground. *Figure 22* shows the cable assembly and the notch in the undertray where clearance was needed for the front wheels when turning.



Figure 22: Front right corner support assembly and notch for steering.

Finally, to reduce the chances of the sharp edges cutting someone, edge trim was added to the front and rear edges. All photos, in chronological order, from the manufacturing process can be found in Appendix F.

5. Testing and Evaluation

Once the undertray manufacturing is complete and finalized, it will be mounted to the car and prepared for dynamic testing. This will help simulate performance in real-world conditions in conjunction with the other sub-systems of the car. A ride height range can then be determined for optimal performance through various operating conditions (e.g. straights and corners). Ride height sensitivity can be determined by 3D printing spacers of various thicknesses that can be mounted between the frame and the undertray, changing the distance between them.

Dynamic testing will also help in better understanding the wake and flow management capacity of the keel section that has been added to the central region of the design. The car will be driven with and without the undertray by the team's drivers; this will allow the team to gauge whether the undertray has had a positive impact on the vehicle's performance and by how much.

The various parameters that can be scrutinized and compared after this test include lap time, maximum lateral G-forces, minimum speed through corners, straight-line speed, and driver confidence. The first four parameters can be measured using the data acquisition instruments provided by Binghamton Motorsports, while the last parameter can be determined by collecting driver feedback post-testing.

Due to complications from other subteams, the EV2 vehicle is currently unable to run. As a result, any real-world testing of the undertray has not been possible, preventing the collection of accurate data. This limitation restricts our ability to validate our design and evaluate the effectiveness of the undertray under operating conditions.

Alternatively, we conducted yarn testing, also known as tuft testing, on the undertray. This is a common practice for many racing applications that spans decades. Yarn testing is where small strands of yarn are taped to components of a car, and air is blown over that region either via driving, or in this case, a pressurized air gun. For the undertray, yarn was taped to the underside of the rear diffuser section, and air was blown from the front of the car. To verify that the undertray is working as intended, the yarn at the rear end of the car must lift and run parallel with the undertray while under, and once the undertray ends, the yarn should lift above it. During our testing, we found exactly what we had seen in CFD simulations. When the air was blown through the front end of the diffuser, the yarn taped to the rear end lifted to level and even above the end of the undertray. This verified that the diffuser is allowing the air to expand as intended, and can be seen in the following figure, *Figure 23*.



Figure 23: Undertray Yarn Testing

6. Schedule and Finances

6.1 Schedule

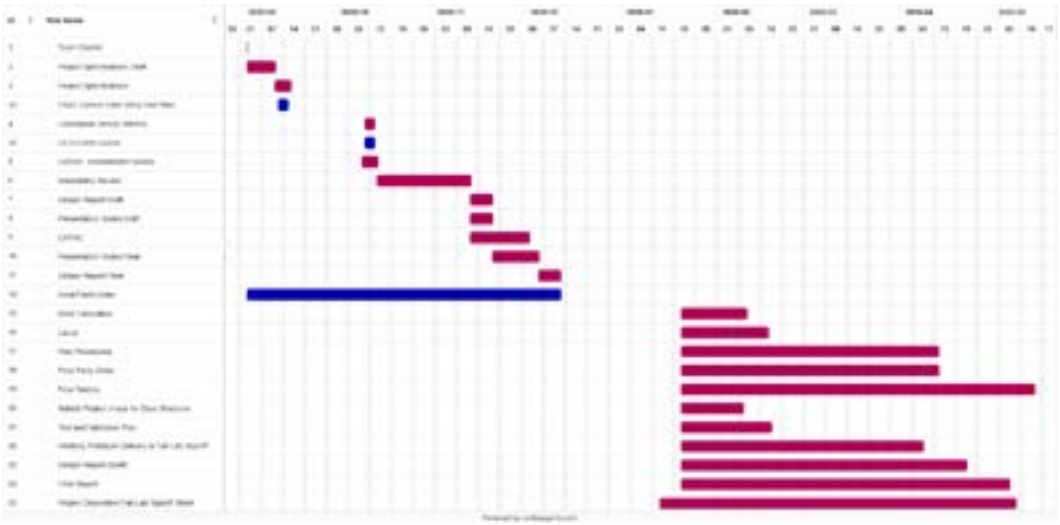


Figure 18: SDP128 Fall through Spring Gantt Chart

6.2 Finances

The following table, *Table 1*, provides a condensed overview of items that have been purchased, followed by their cost rounded to the nearest dollar. See Appendix A for the full Bill of Materials. Note that several items were donated by the Binghamton Motorsports FSAE Aero team, including 12k carbon fiber, sandpaper, and wooden dowels. Additionally, no ½ inch foam insulation board was necessary for project completion and therefore was not purchased.

Table 1: Current cost-tracking table.

Item	Original Estimate ($)	Actual to Date ($)	Estimate to Completion ($)	Estimate at Completion ($)
Carbon fiber fabric (3k @ 50”×10 yd)	108	550	0	550
Carbon fiber fabric (12k @ 50”×5 yd)	191	0	0	0
Epoxy, hardener, and bottle pumps	450	386	0	386
½” Foam Insulation Board	19	0	0	0
1” Foam Insulation Board	32	64	0	64
2” Foam Insulation Board	63	63	0	63
3D printer Filament (PLA)	21	22	0	22
Drywall Plaster (1 Gal)	15	26	0	26
PVA Release Film	15	30	0	0
Paint Brushes	8	30	0	30
Peel Ply & Bleeder Cloth	46	208	0	208
Vacuum Bag & Putty Sealant	18	38	0	38
Sandpaper Kit (180-2000 grit)	17	0	0	0
Wood Dowels (1”×1”×36”)	6	0	0	0
Total Spent:	**1,387**			
Remaining Budget:	**613**			

7. Social Impact

The “social impact” of SDP128’s undertray design relates to the members, faculty advisors, and sponsors of Binghamton Motorsports. Binghamton Motorsports Formula SAE team consists of 98 members, various faculty members and alumni serving as advisors, and dozens of sponsors. The total time commitment of the team is so immense that quantifying it is impossible, but each vehicle, from the start of research and design to the final adjustments made at competition, takes tens of thousands of hours. Additionally, the financial commitment made by the university and sponsors is in the tens of thousands of dollars. That being said, every component of the vehicle needs to be perfect, as design and manufacturing flaws can lead to point deductions and potential disqualification. Throughout this project, the team has been conscientious of the work they complete. The previous undertray, although used on two vehicles at competition, was not rule-compliant for various reasons. Jeopardizing the time and money sacrificed by hundreds of people is not an option for SDP128.

Furthermore, the design of the undertray was developed with careful consideration of user safety and overall system reliability. By improving vehicle stability and aerodynamic performance, the design supports safer vehicle operation, especially at high speeds. Additionally, the structural strength of the part as well as its material composition were carefully selected to withstand the operating forces without breaking down. This approach helps minimize potential risks while maintaining reliable and consistent performance.

In addition to safety, environmental, economic, and social factors were taken into account throughout the development process. The weight of materials used in the design was compared with the goal of optimizing the energy use and minimizing the amount of raw materials required for fabrication. The relationship between cost and performance was also assessed to ensure that the design remains practical within budget constraints. Overall, the project illustrates a good balance and compromise between performance, sustainability, and cost, while demonstrating awareness of broader social impacts.

8. References

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Ooq48UiOnXi3FUTN1tO1pmdLwO1fof-G8QiW5HgQxHh573AgByjT (accessed Nov. 21, 2025).

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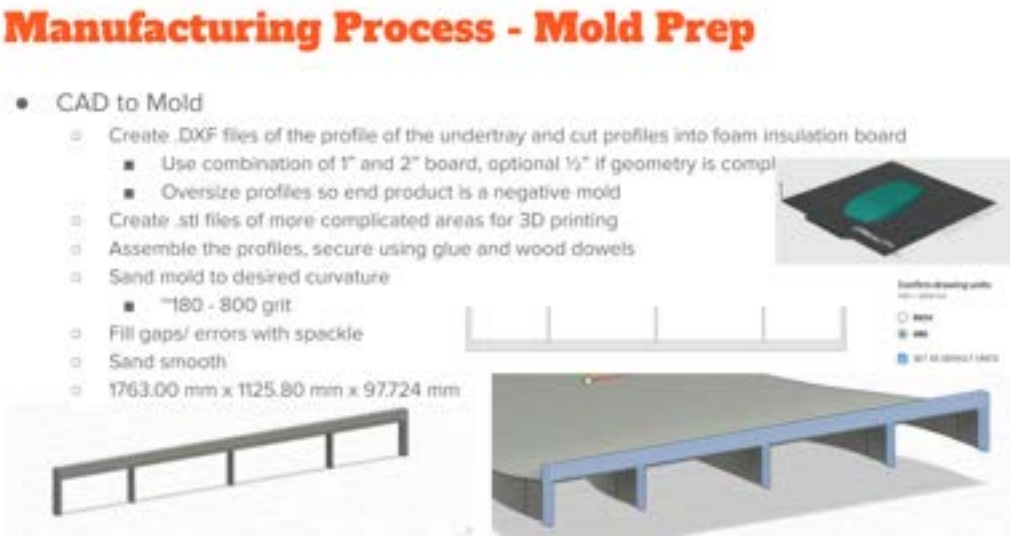
[8] O. H. Ehirim, K. Knowles, and A. J. Saddington, "A Review of Ground-Effect Diffuser Aerodynamics," Journal of Fluids Engineering, vol. 141, no. 2, Feb. 2019.

doi: 10.1115/1.4040501.

Appendices

Appendix A: Design Makeability Review





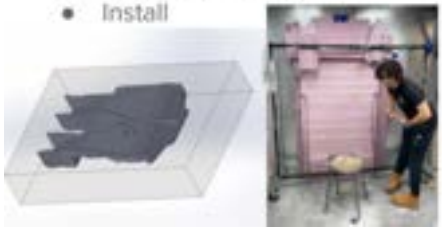
Manufacturing Process - Carbon Layup

- Coat mold with PVA release agent
- Cut carbon fiber sheets to size
  - Test fit to determine areas that will require more focus
- Gather peel ply, breather cloth, vacuum bag, vacuum pump
- Prepare Epoxy/hardener mixture (Slow hardener to allow more working time)
- Layup
  - Layer A of carbon, Layer A of epoxy
  - Layer B of carbon, Layer B of epoxy
  - Layer C of carbon, Layer C of epoxy
  - Peel ply sheet
  - Breather cloth
  - Vacuum bag



Manufacturing Process - Post Processing

- Remove Vacuum bag, peel away peel ply & breather cloth
- Remove composite from foam mold
- Trim edges to size
- Wet sand both sides from ~180 - 2000 grit
- Sand edges smooth for rule compliance
- Cut into sections
- Install







Material	Quantity	Price
½", 1", 2" foam insulation boards	TBD	$18.97, $31.97, $62.97
3D printer filament (PLA, 1 kg spool)	1 spool	$20.66
Sandpaper kit (180–800 grit, mold prep)	1 kit	$15.07
Drywall plaster (1 quart)	1 unit	$10.86
Wood dowels (1"×1"×36")	9 pieces	$51.88
PVA release film	1 set	$14.51
Carbon fiber fabric (3k @ 50"×3 yd + 12k @ 50"×1 yd)	2 rolls	($108.10, $190.50)
Epoxy, hardener, and bottle pumps	1 set	$450.19
Paint brushes	1 set	$7.62
Peel ply cloth	1 sheet	$23.73
Bleeder cloth	1 sheet	$22.35
Vacuum bag	1 bag	$7.83
Vacuum bag putty sealant	1 pack	$10.07

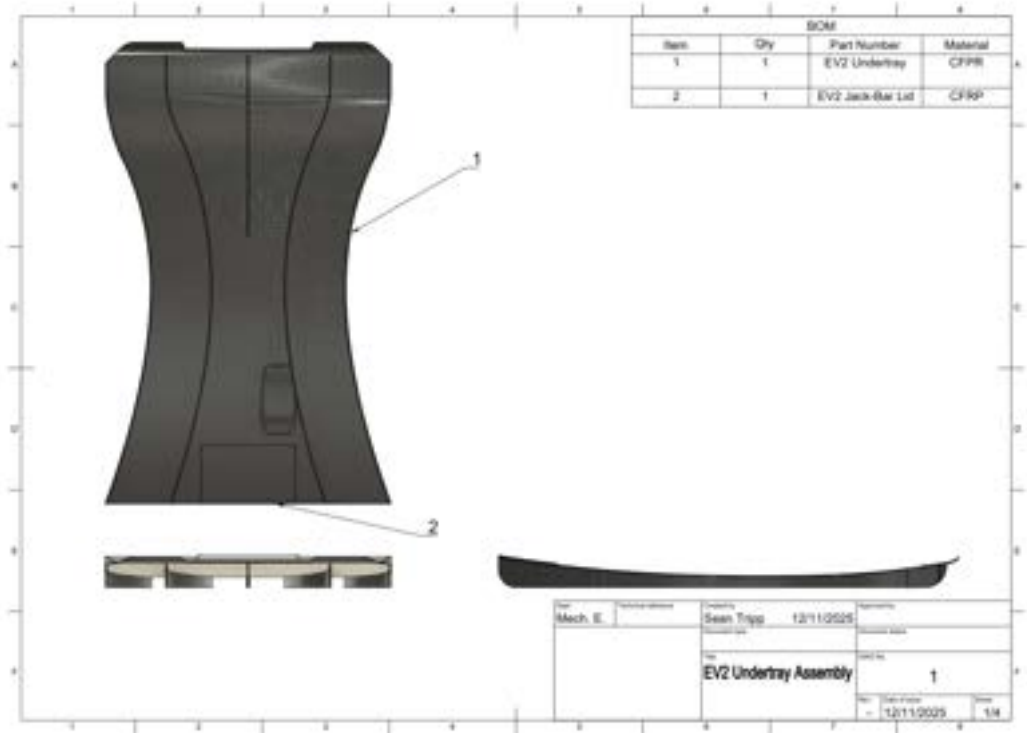
Bill of Materials

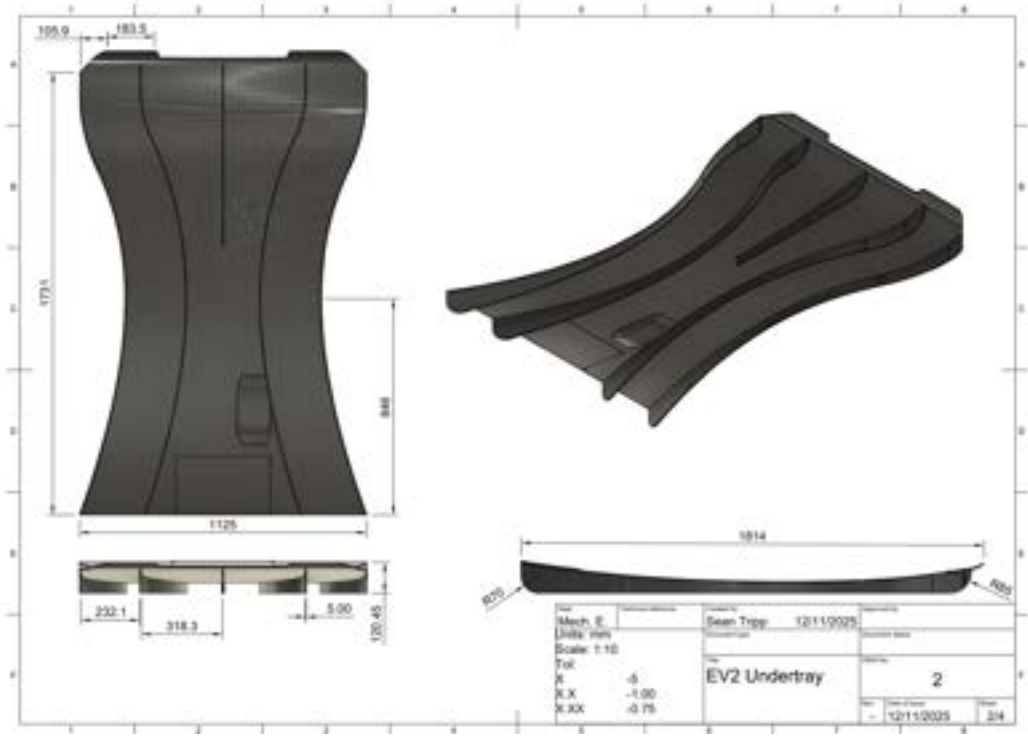
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  - Total Boat
  - EasyComposites
  - Chomarat
  - Home Depot
  - Lowes
  - Harbor Freight

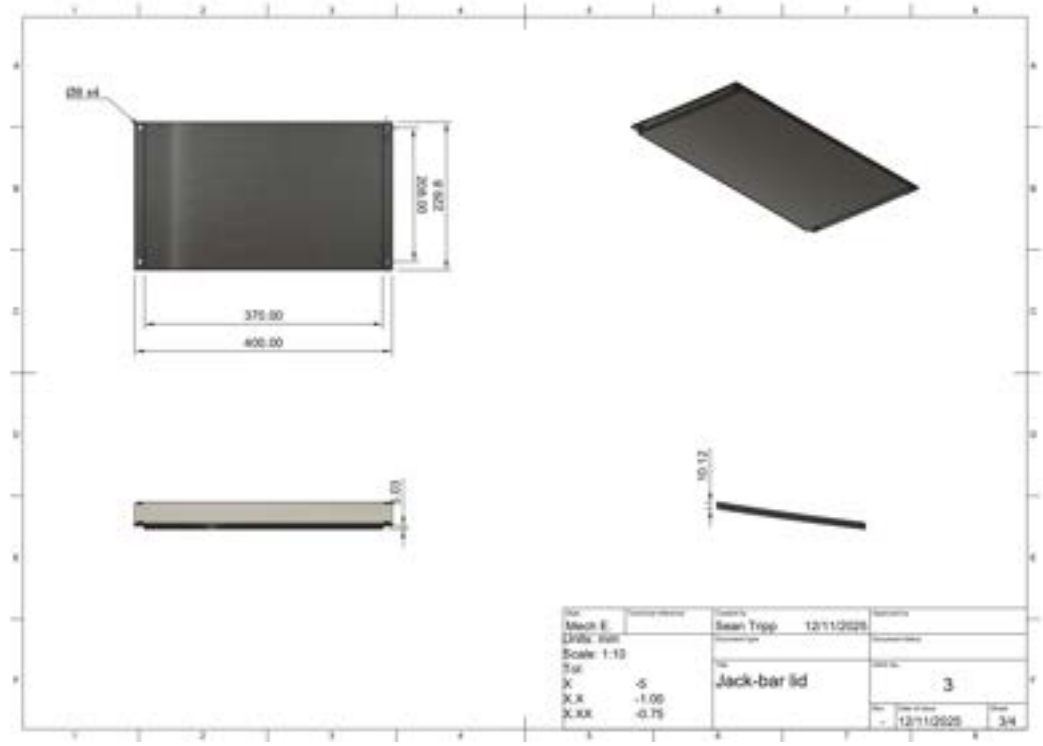
Appendix B: Bill of Materials (BOM)

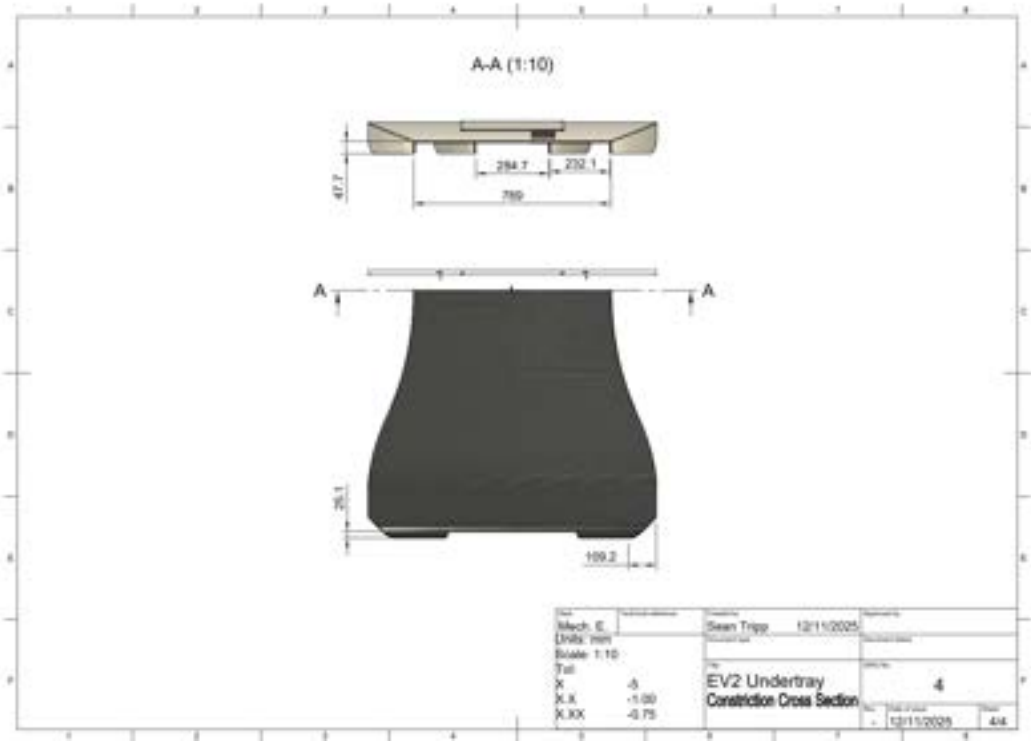
Material	Quantity	Price
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Drywall plaster (1 quart)	1 unit	$10.86
Wood dowels (1”×1”×36”)	9 pieces	$51.88
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Epoxy, hardener, and bottle pumps	1 set	$450.19
Paint brushes	1 set	$7.62
Peel ply cloth	1 sheet	$23.73
Bleeder cloth	1 sheet	$22.35
Vacuum bag	1 bag	$7.83
Vacuum bag putty sealant	1 pack	$10.07
Sandpaper kit (180–2000 grit, post-processing)	1 kit	$17.39

Appendix C: Detailed CAD Drawings



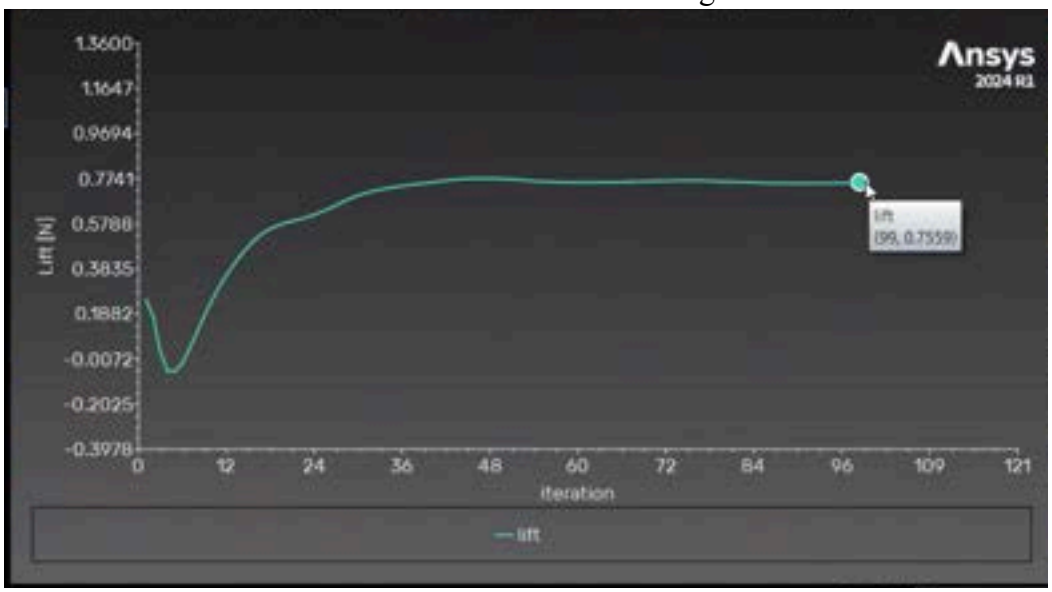



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400.00
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Units: mm
Scale: 1:10
Tol:
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Sean Tripp 12/11/2025
Jack-bar lid
3
12/11/2025
3/4

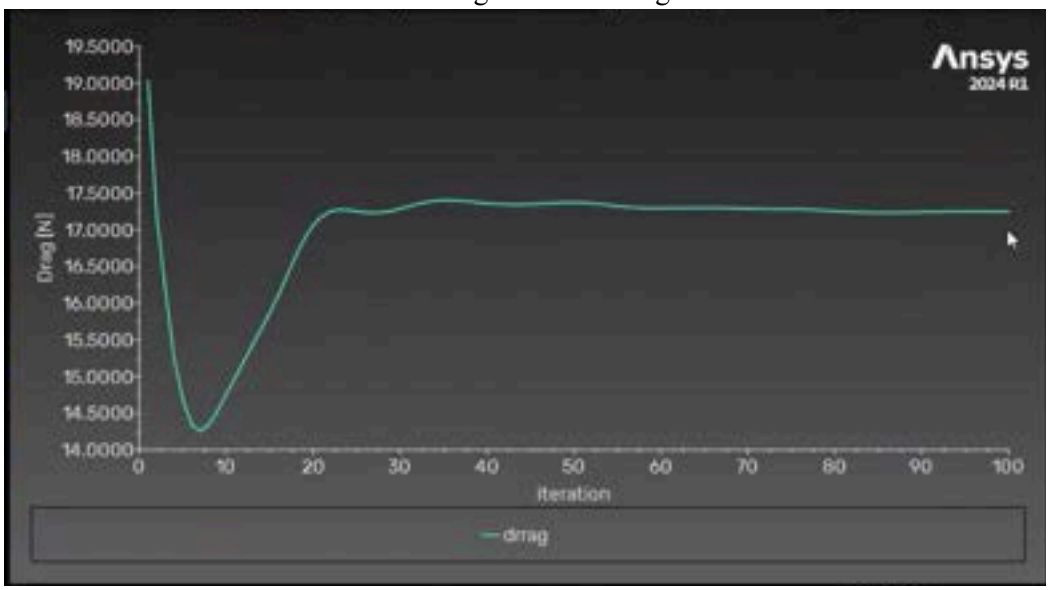

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X -5
X.X -1.00
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EV2 Undertray
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12/11/2025
4/4

Appendix D: CFD results

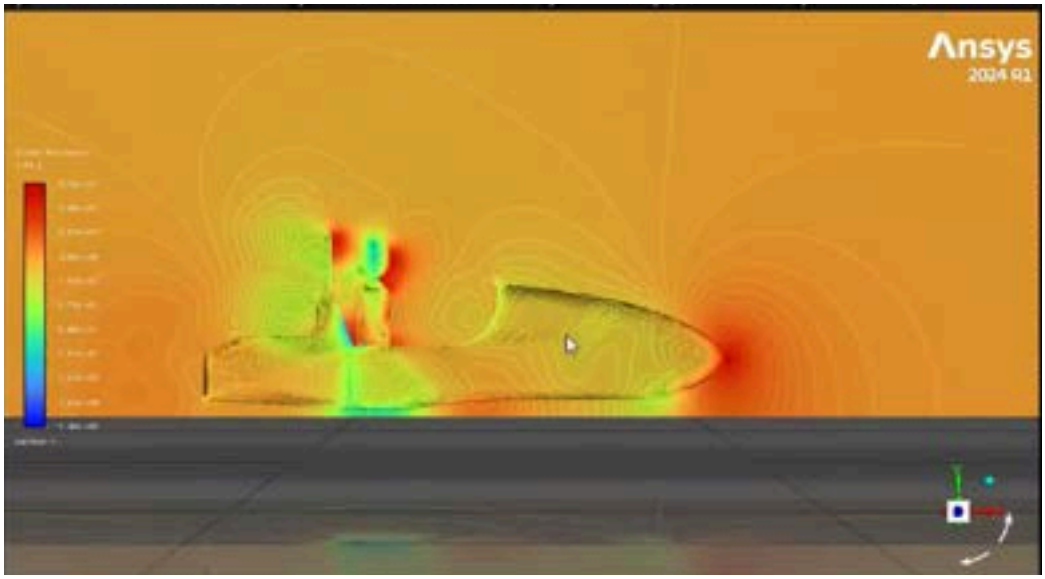
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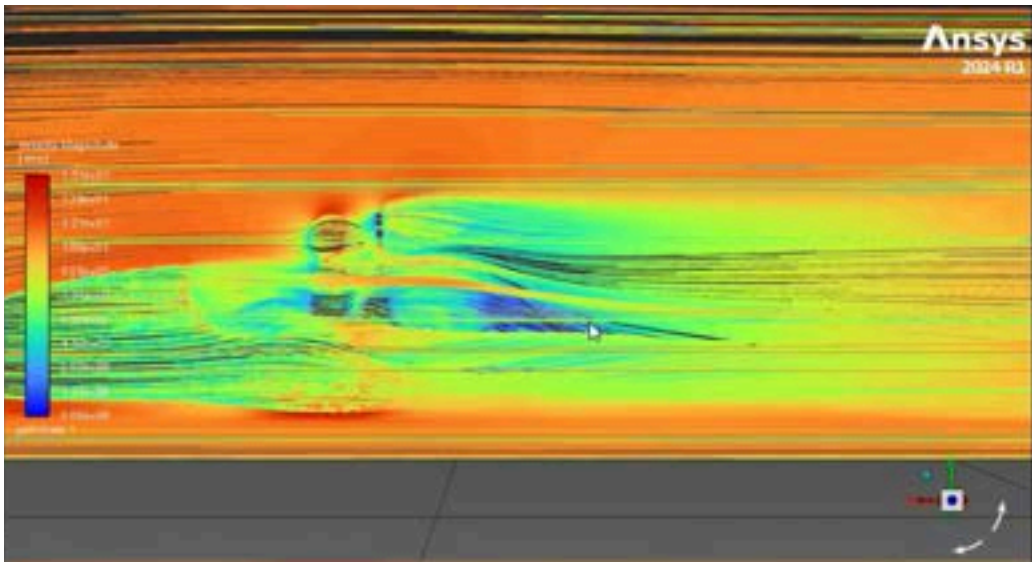
Baseline Black-box - CFD Results - Drag Force Convergence



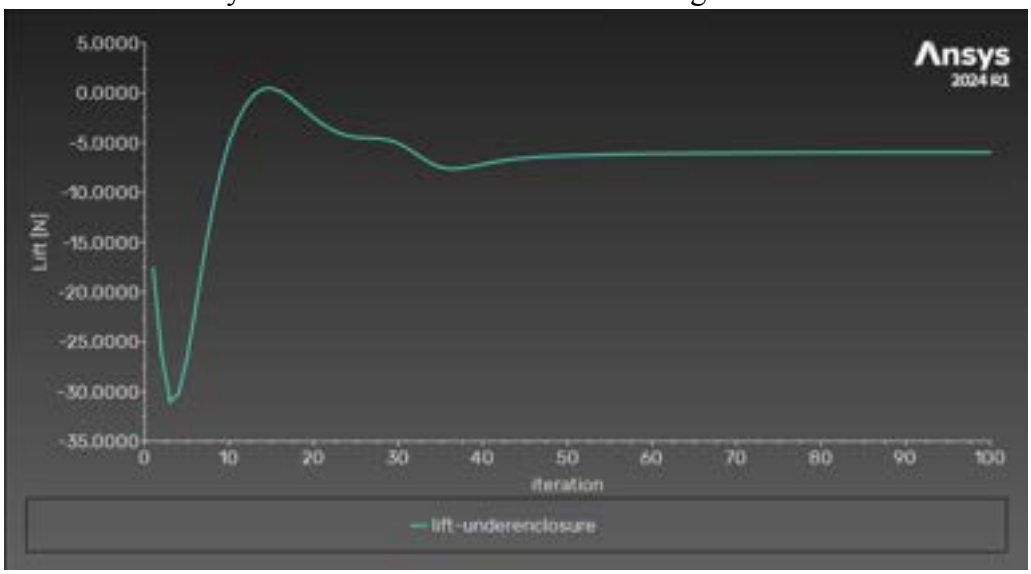
Baseline Black-box - CFD Results - Side View Pressure Contour Plot



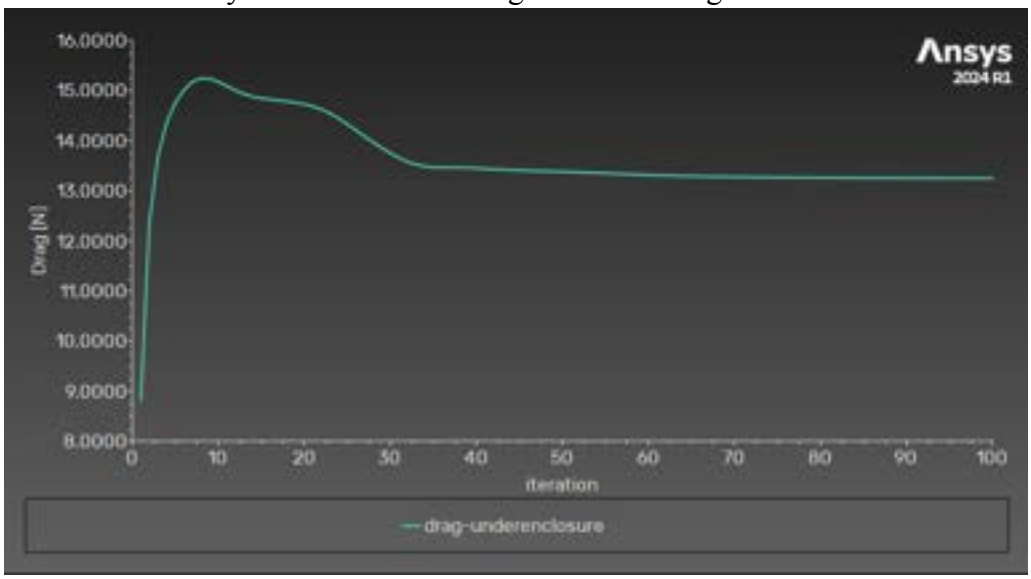
Baseline Black-box - CFD Results - Side View Pathline Plot



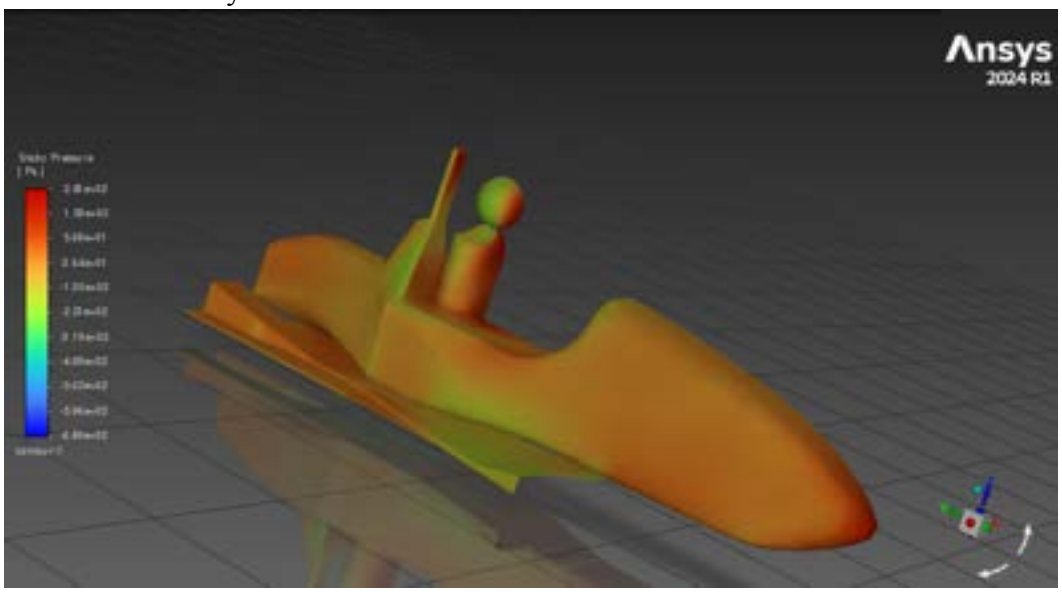
Previous undertray - CFD Results - Lift Force Convergence



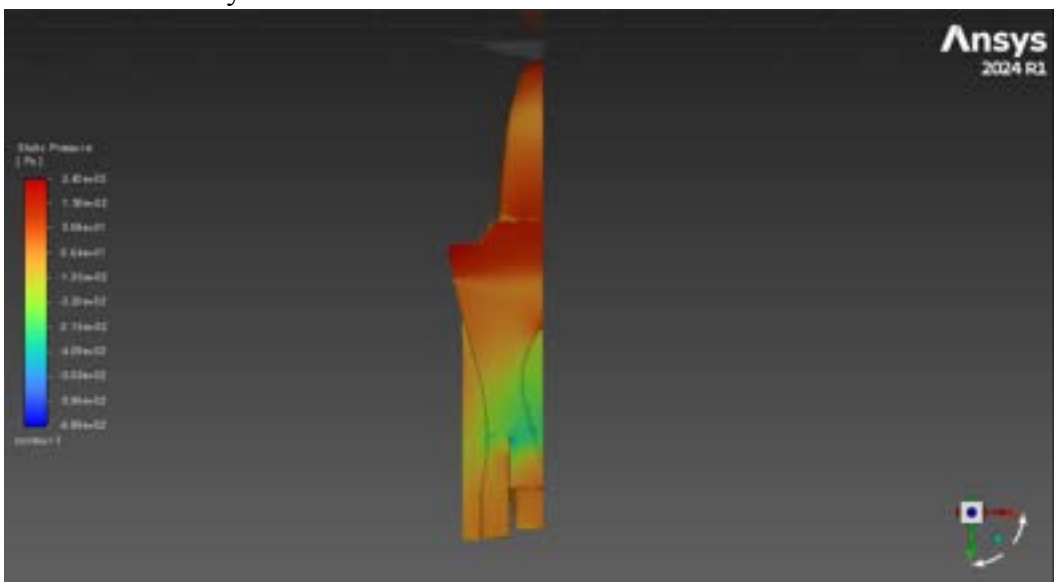
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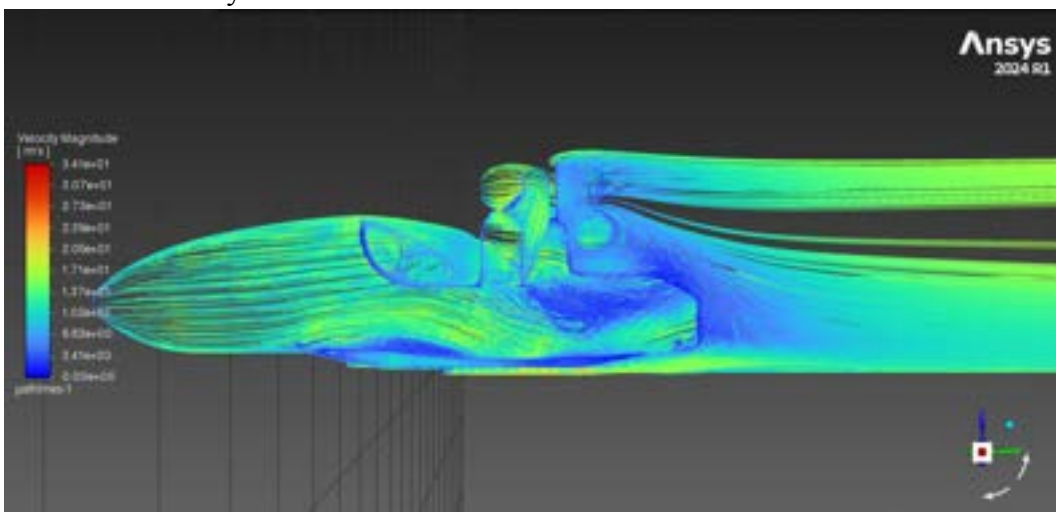
Previous undertray - CFD Results - Isometric View Pressure Contour Plot



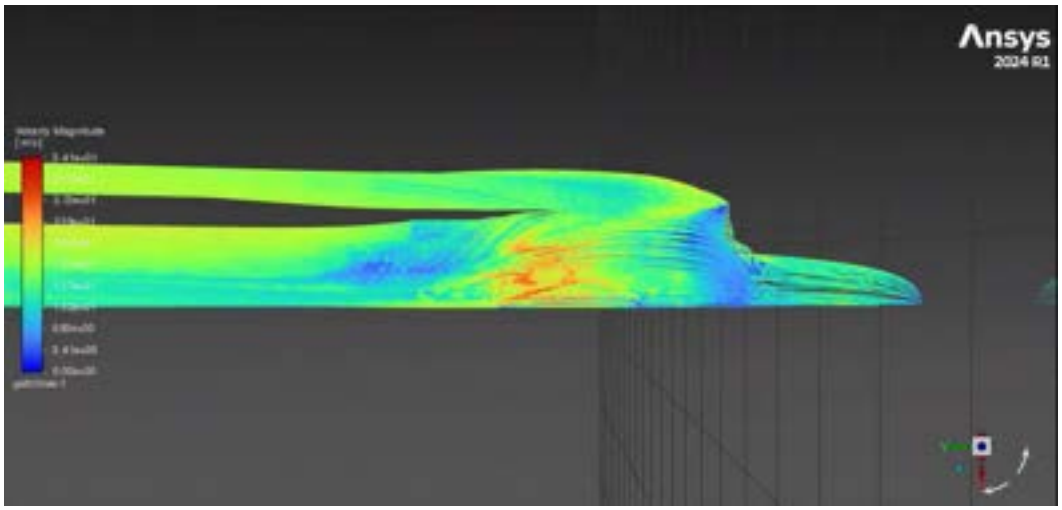
Previous undertray - CFD Results - Bottom View Pressure Contour Plot



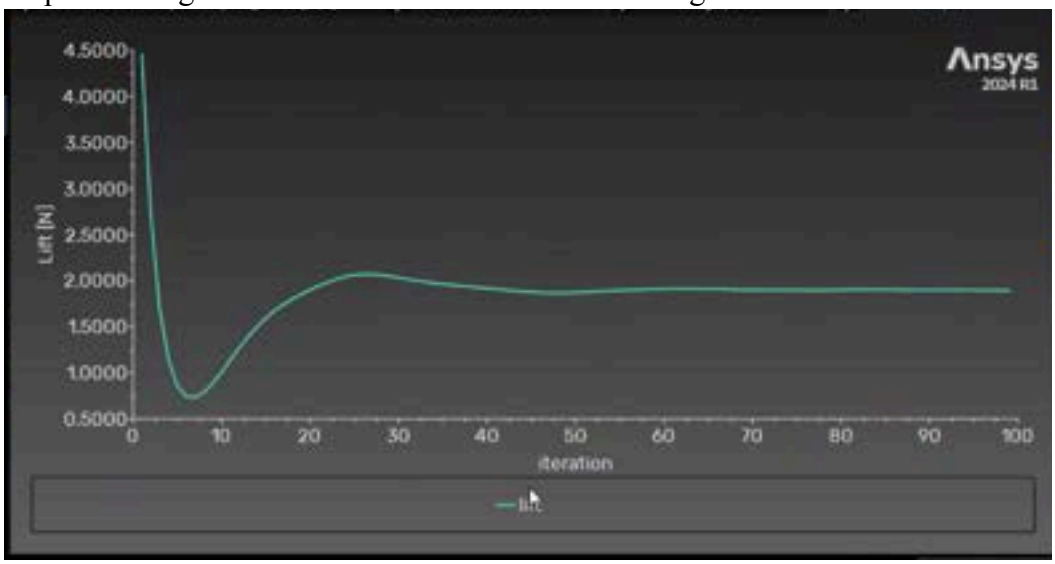
Previous undertray - CFD Results - Side View Pathline Plot



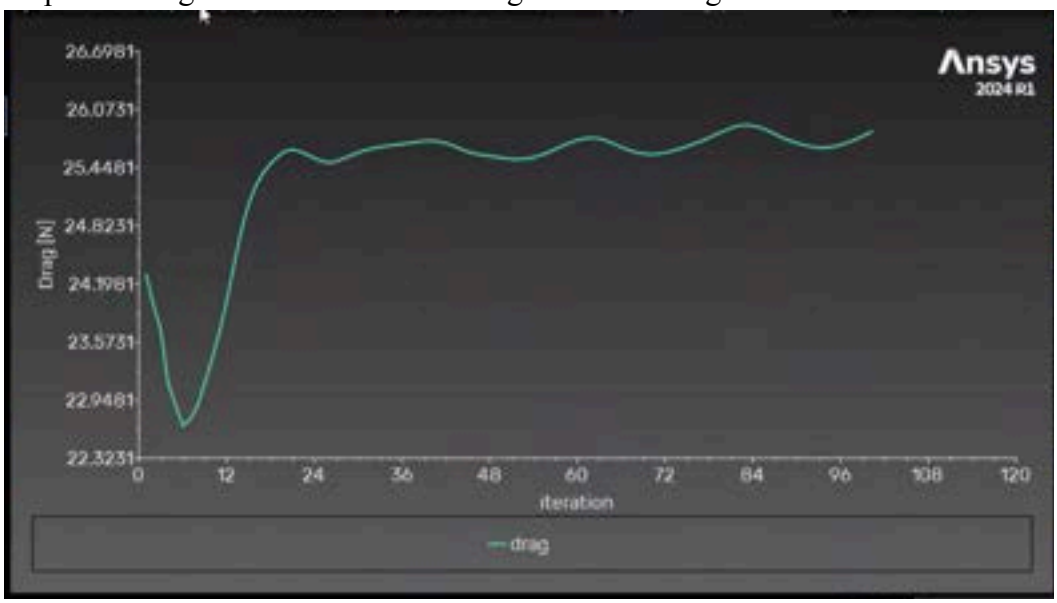
Previous undertray - CFD Results - Bottom View Pathline Plot



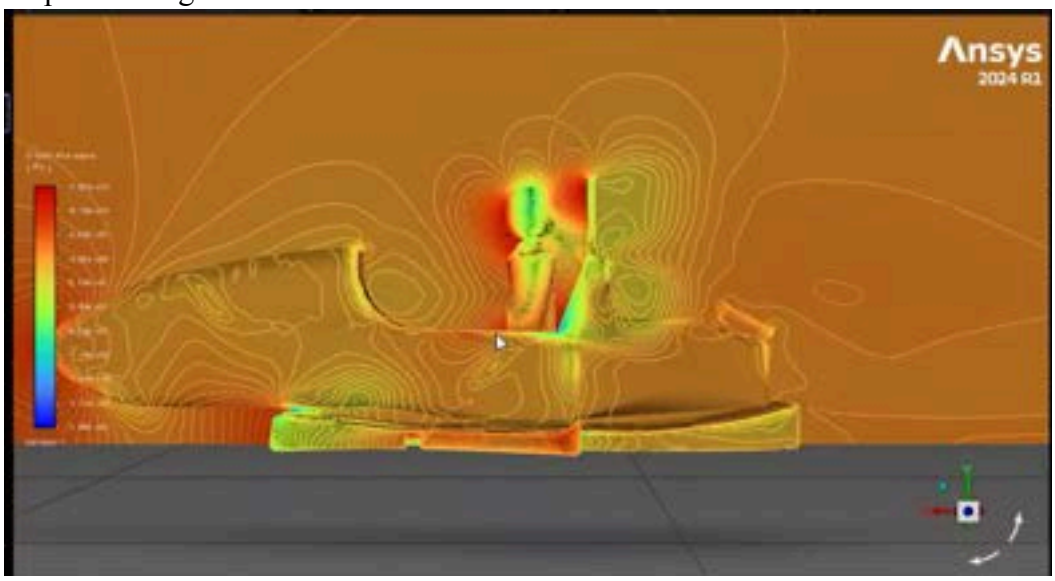
Proposed design A - CFD Results - Lift Force Convergence



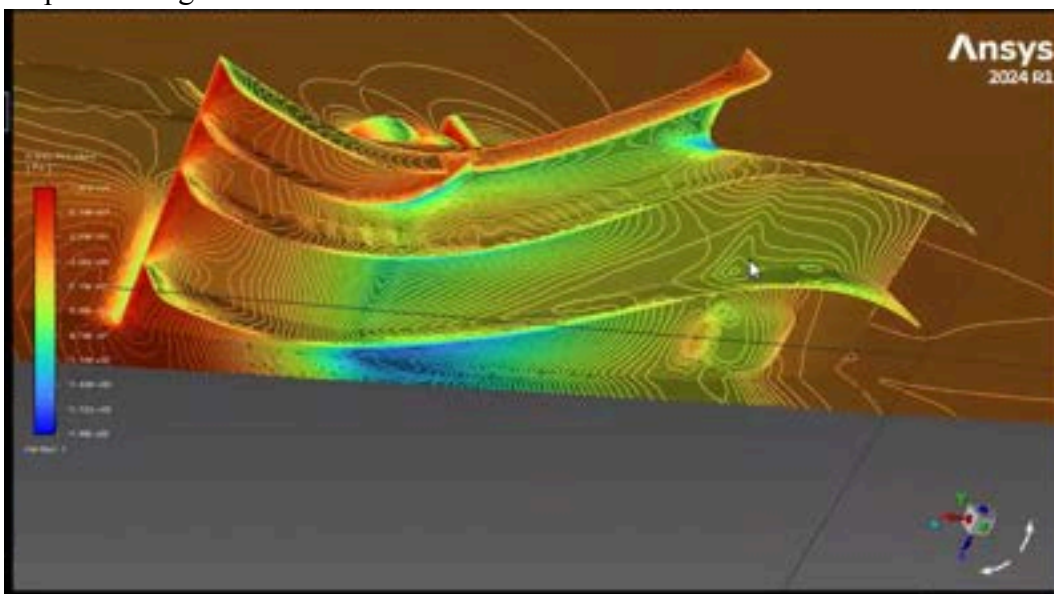
Proposed design A - CFD Results - Drag Force Convergence



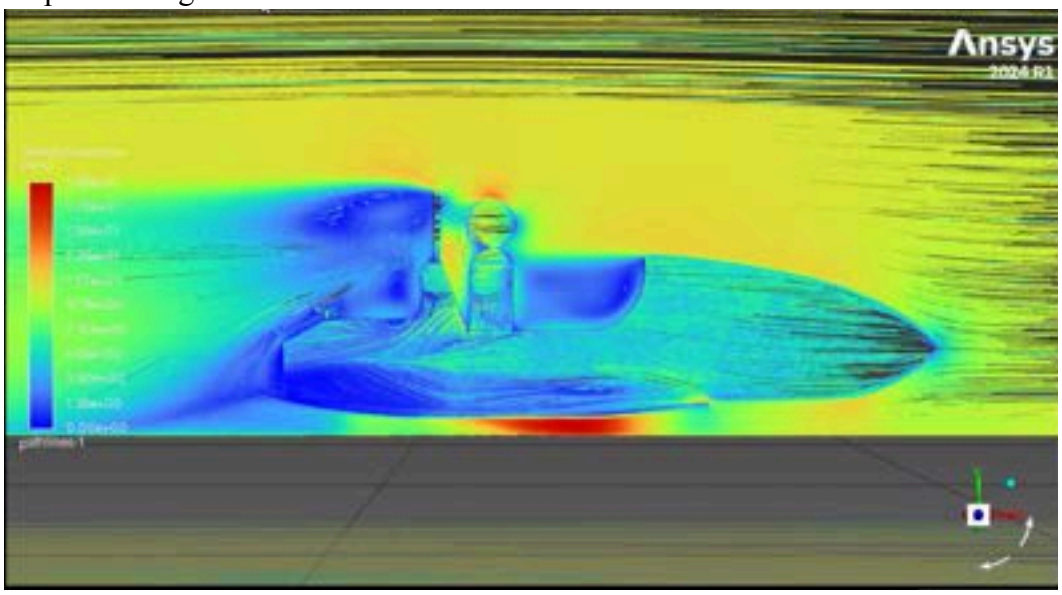
Proposed design A - CFD Results - Side View Pressure Contour Plot



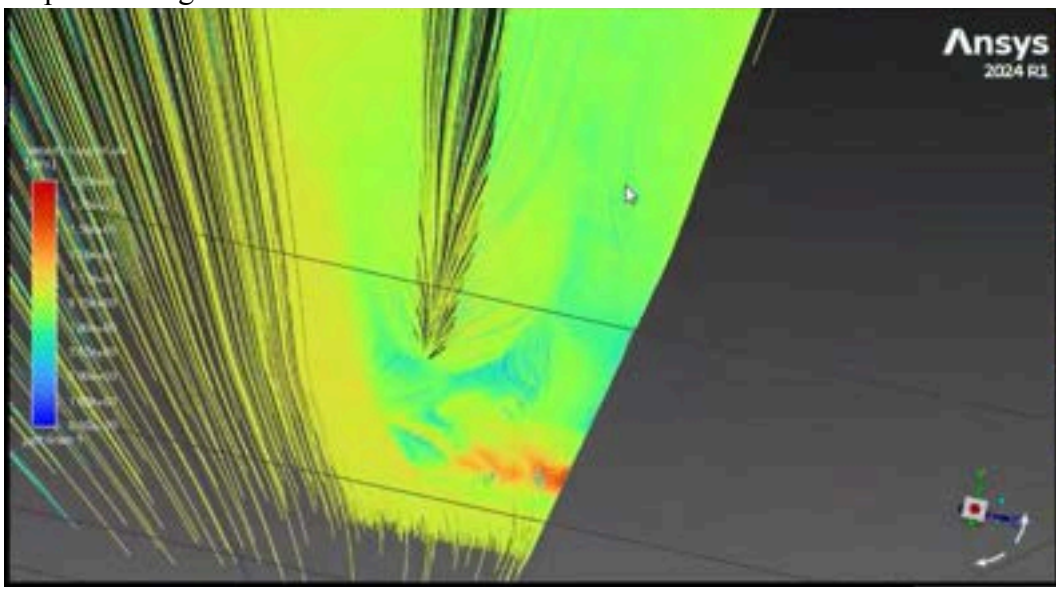
Proposed design A - CFD Results - Bottom View Pressure Contour Plot



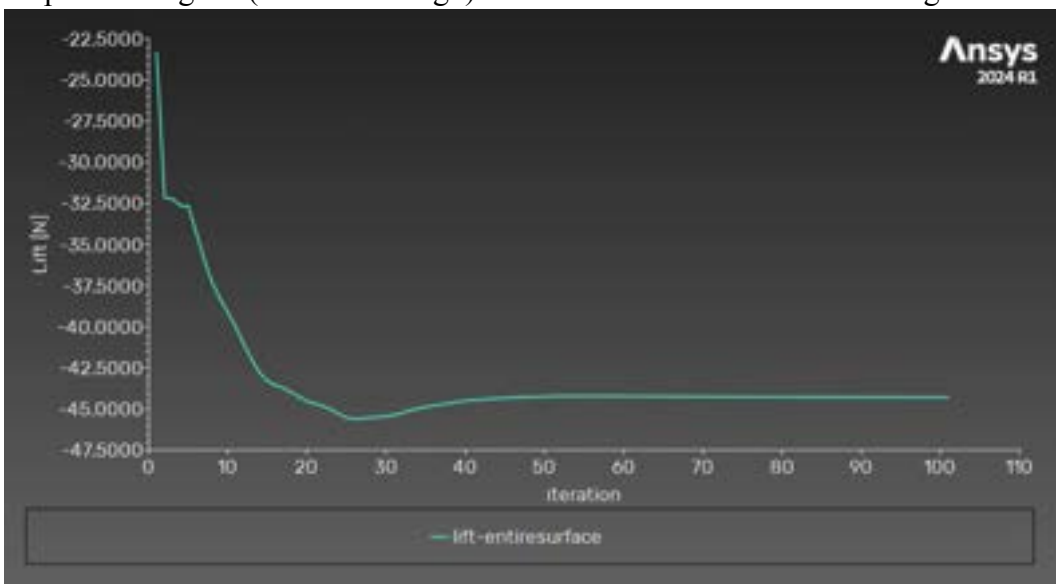
Proposed design A - CFD Results - Side View Pathline Plot



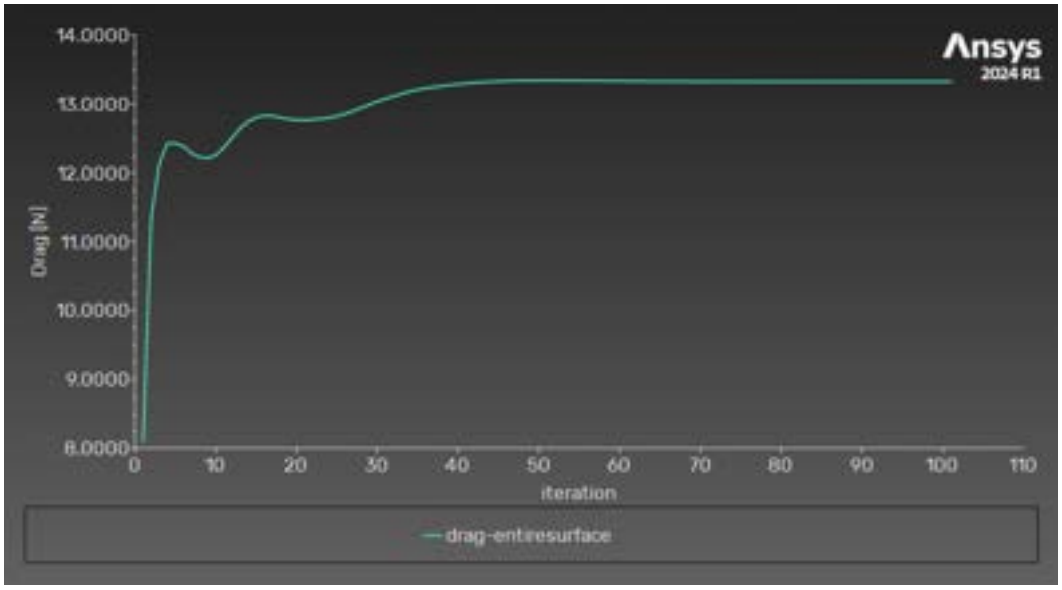
Proposed design A - CFD Results - Bottom View Pathline Plot



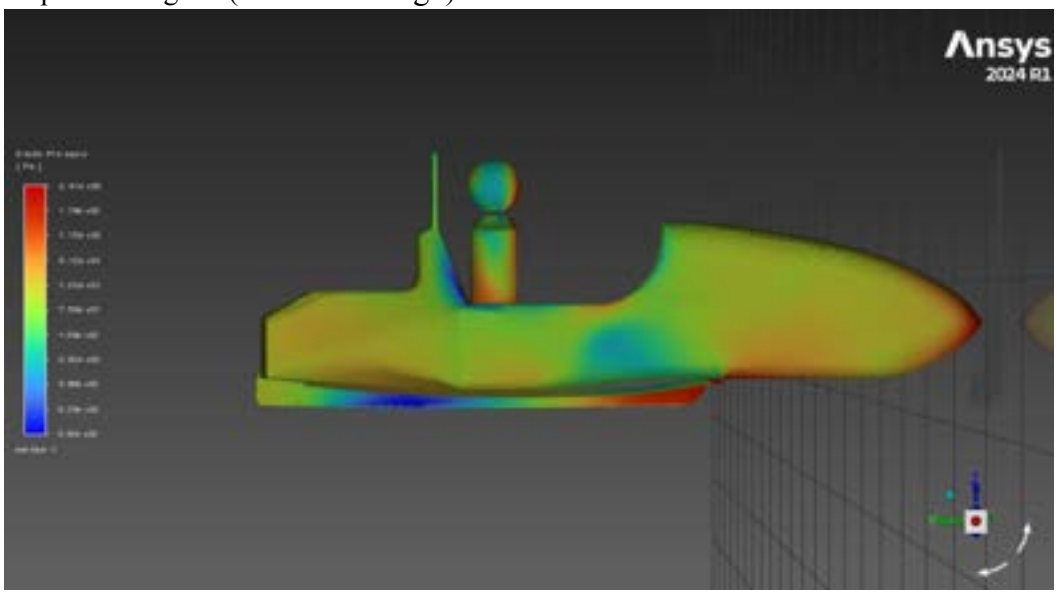
Proposed design B (Selected Design) - CFD Results - Lift Force Convergence



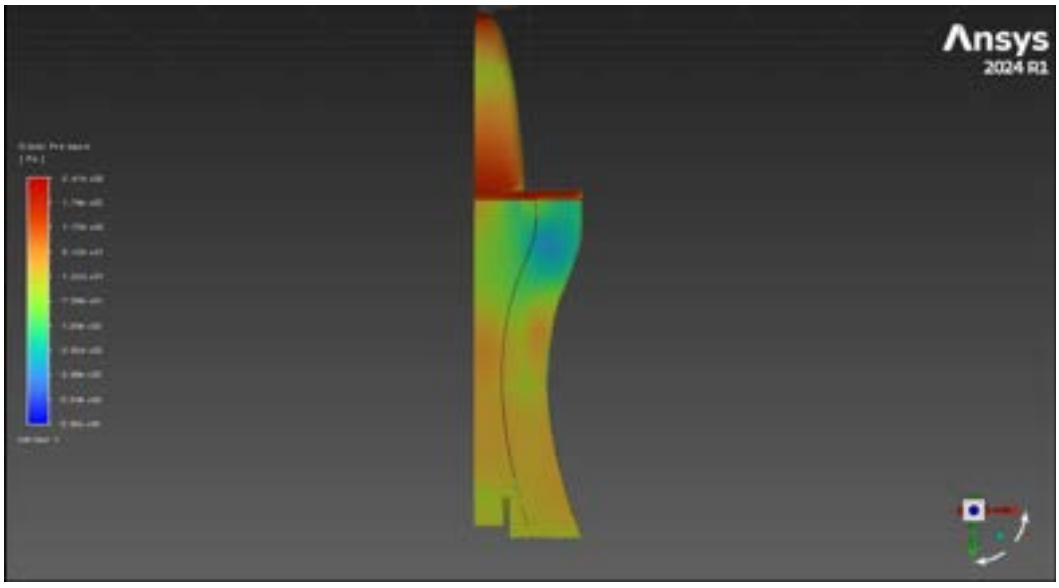
Proposed design B (Selected Design) - CFD Results - Drag Force Convergence



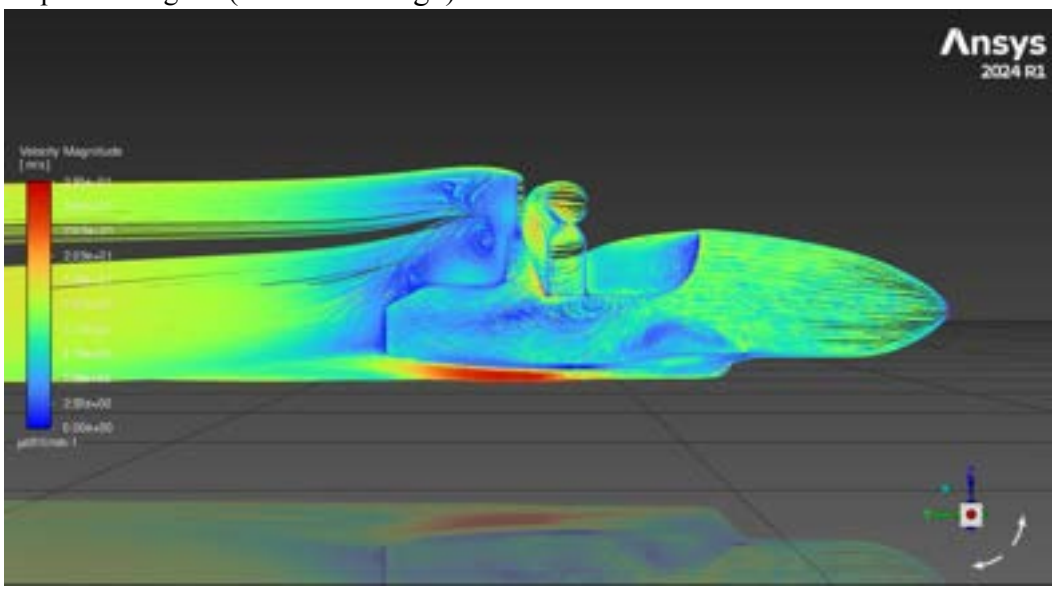
Proposed design B (Selected Design) - CFD Results - Side View Pressure Contour Plot



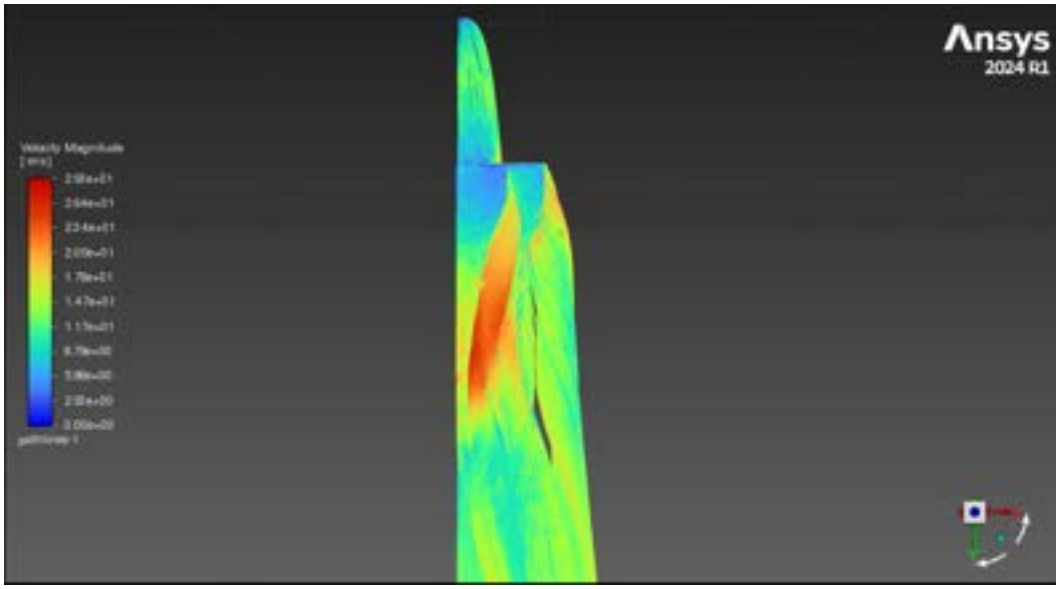
Proposed design B (Selected Design) - CFD Results - Bottom View Pressure Contour Plot



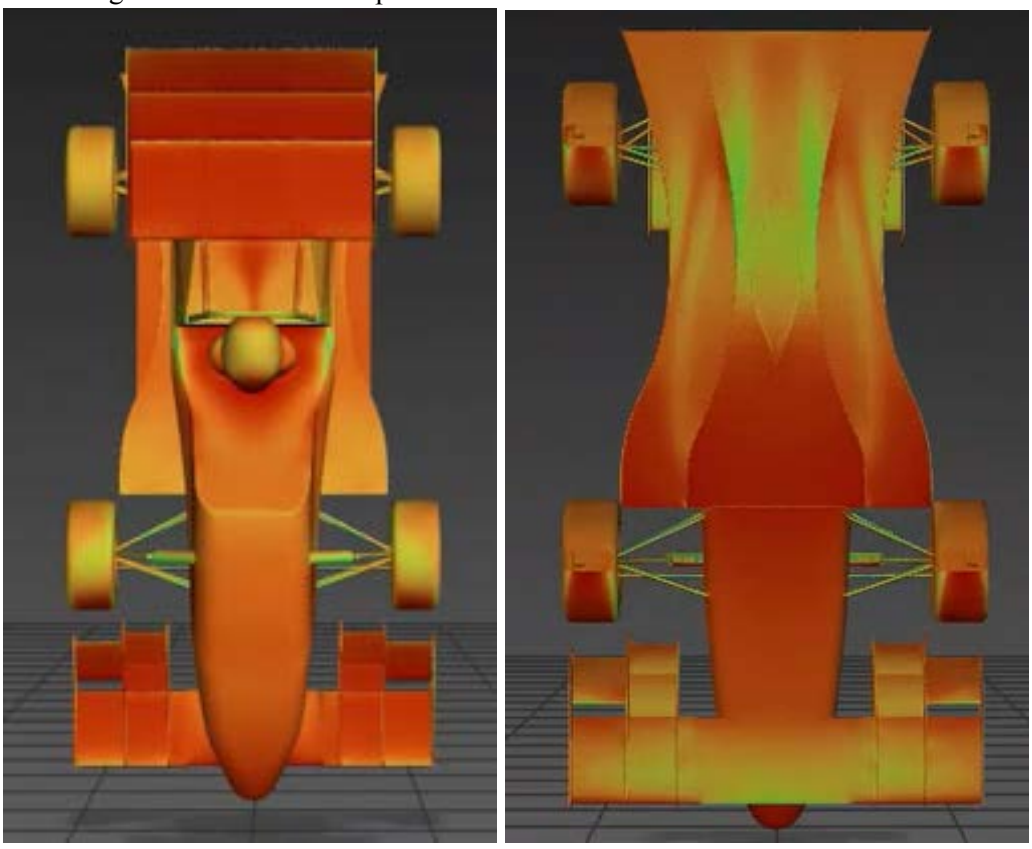
Proposed design B (Selected Design) - CFD Results - Side View Pathline Plot



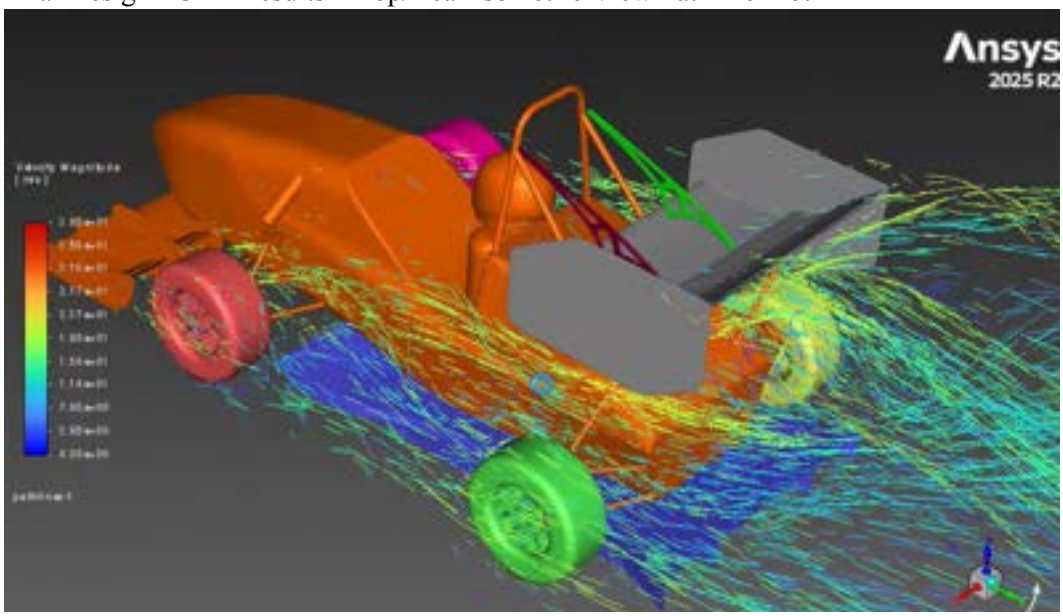
Proposed design B (Selected Design) - CFD Results - Bottom View Pathline Plot



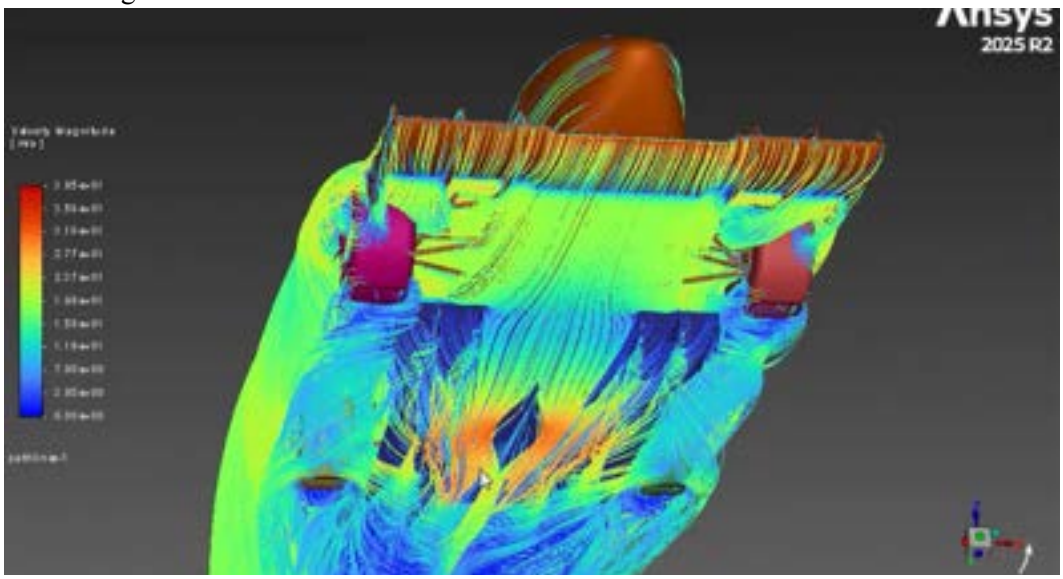
Final Design - CFD Results - Top and Bottom View Contour Plot



Final Design - CFD Results - Top/Rear Isometric View Pathline Plot



Final Design - CFD Results - Bottom/Front Isometric View Pathline Plot



Appendix E: Rejected concepts

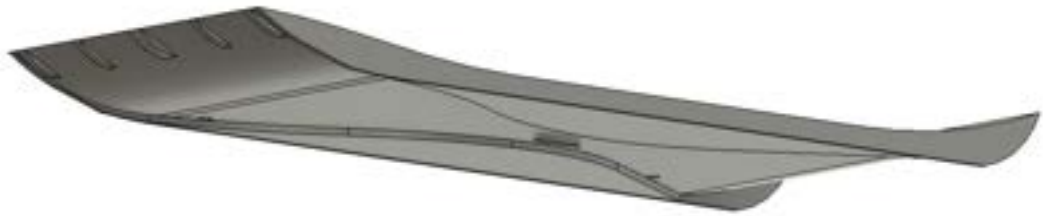
Rejected Concept A - Topside - Front View



Rejected Concept A - Topside - Back View



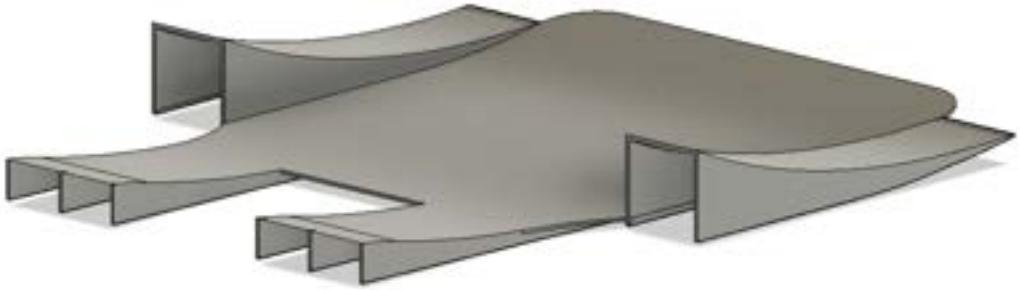
Rejected Concept A - Underside - Front View



Rejected Concept B - Topside - Front View



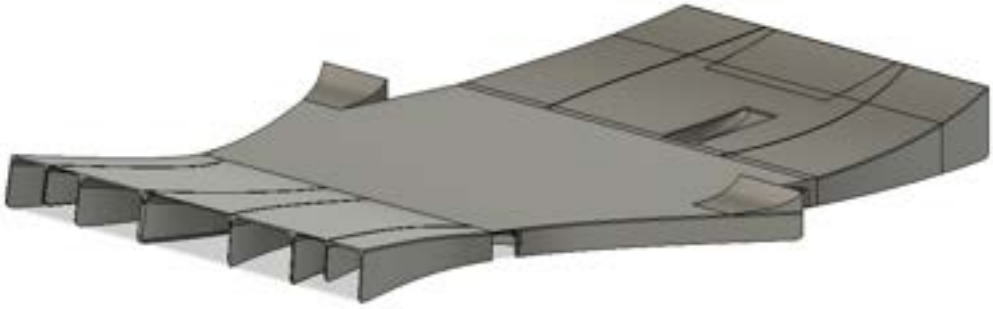
Rejected Concept B - Topside - Back View



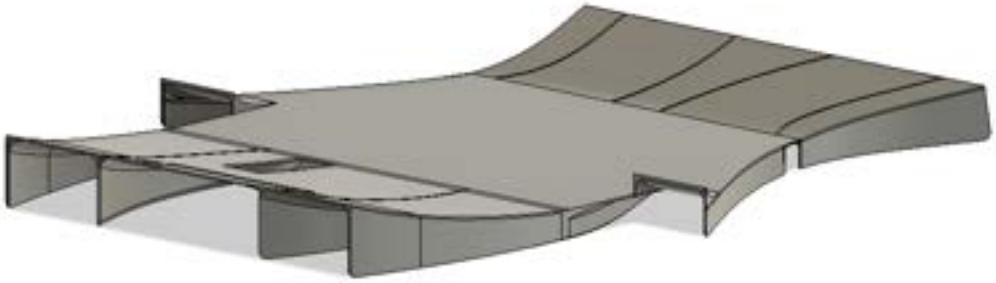
Rejected Concept B - Underside - Front View



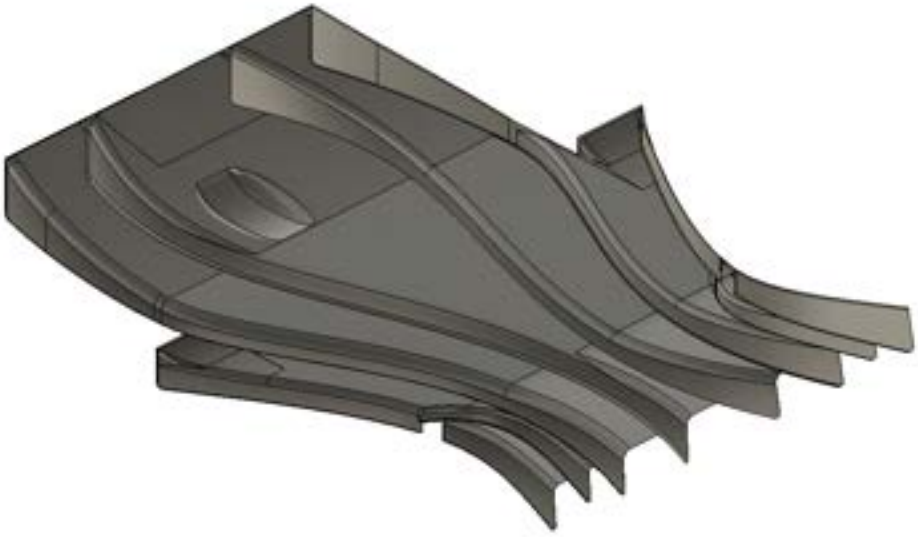
Rejected Concept C (Proposed Design A) - Topside - Front View



Rejected Concept C (Proposed Design A) - Topside - Back View



Rejected Concept C (Proposed Design A) - Underside - Back View



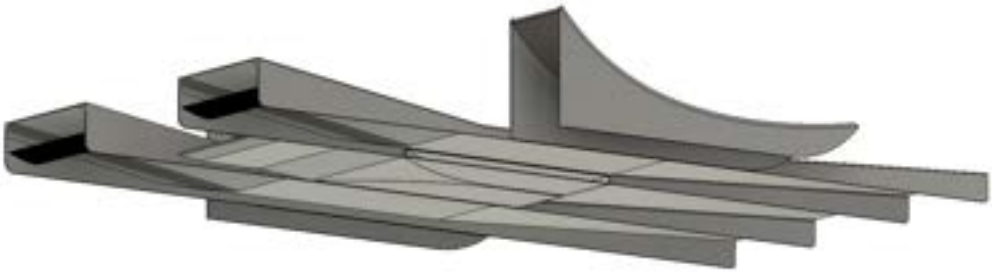
Rejected Concept D - Topside - Front View



Rejected Concept D - Topside - Back View

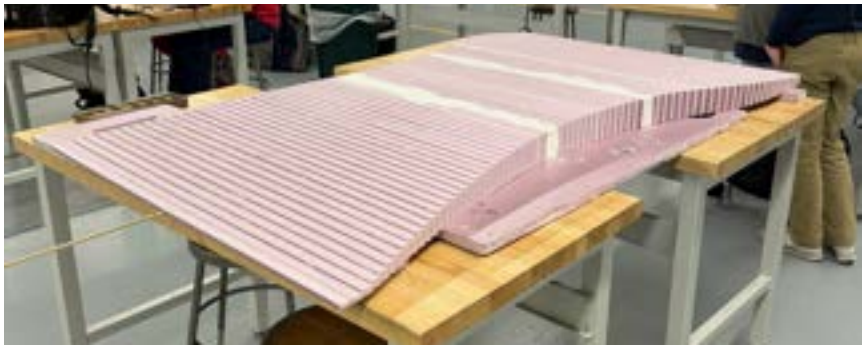


Rejected Concept D - Underside - Back View



Appendix F: Manufacturing Process Photos

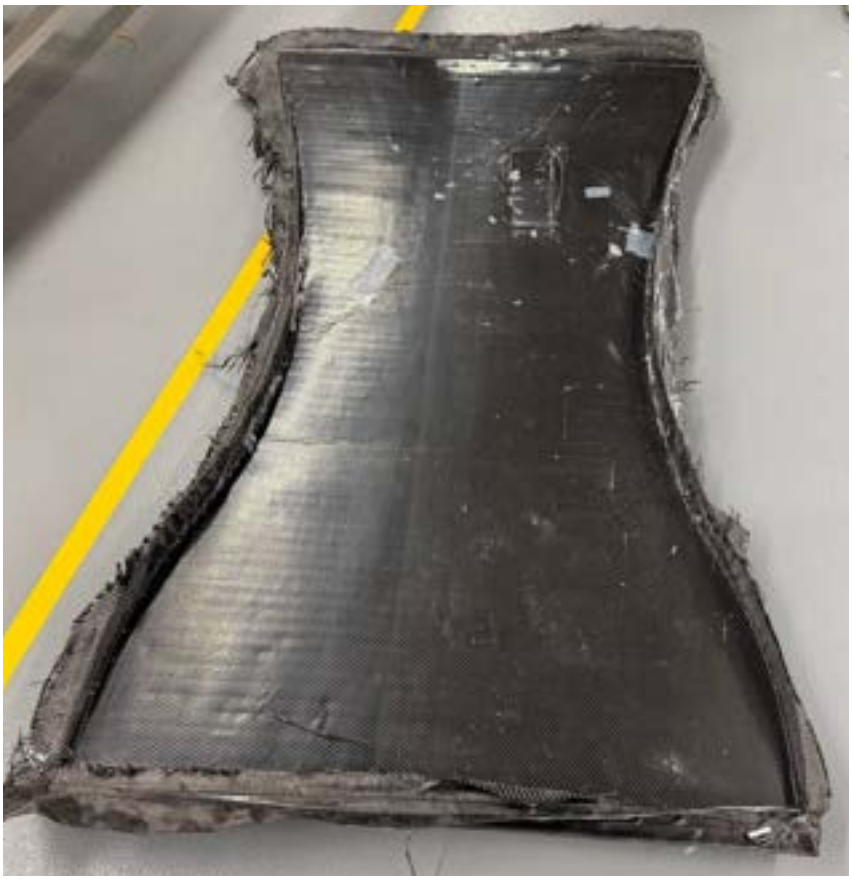
Final undertray foam mold, pre-sanding:



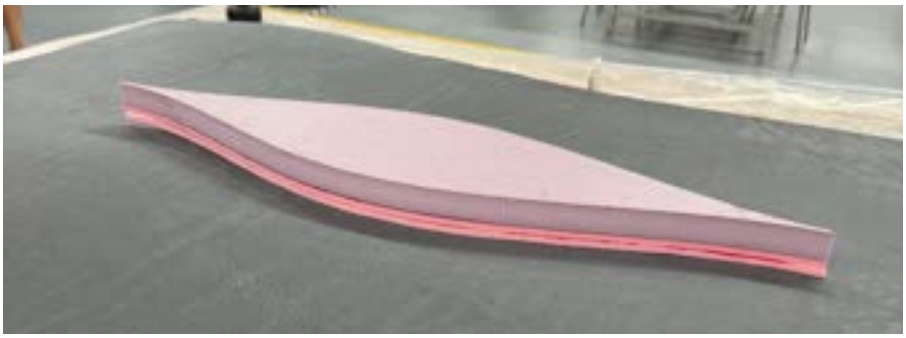
Underside of undertray after initial layup:



Topside of undertray after initial layup:



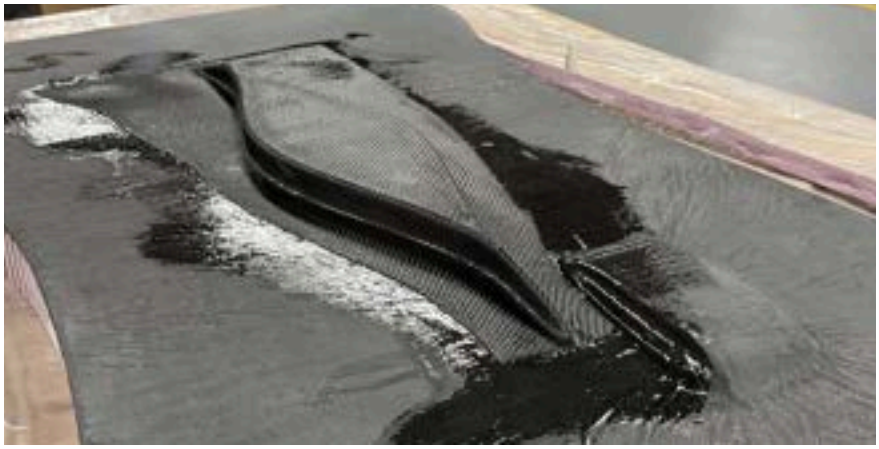
Foam re-energizer with 1x 1-inch and 2x ¼-inch sections:



Re-energizer during prep before layup:



Re-energizer immediately after being epoxied to the undertray:



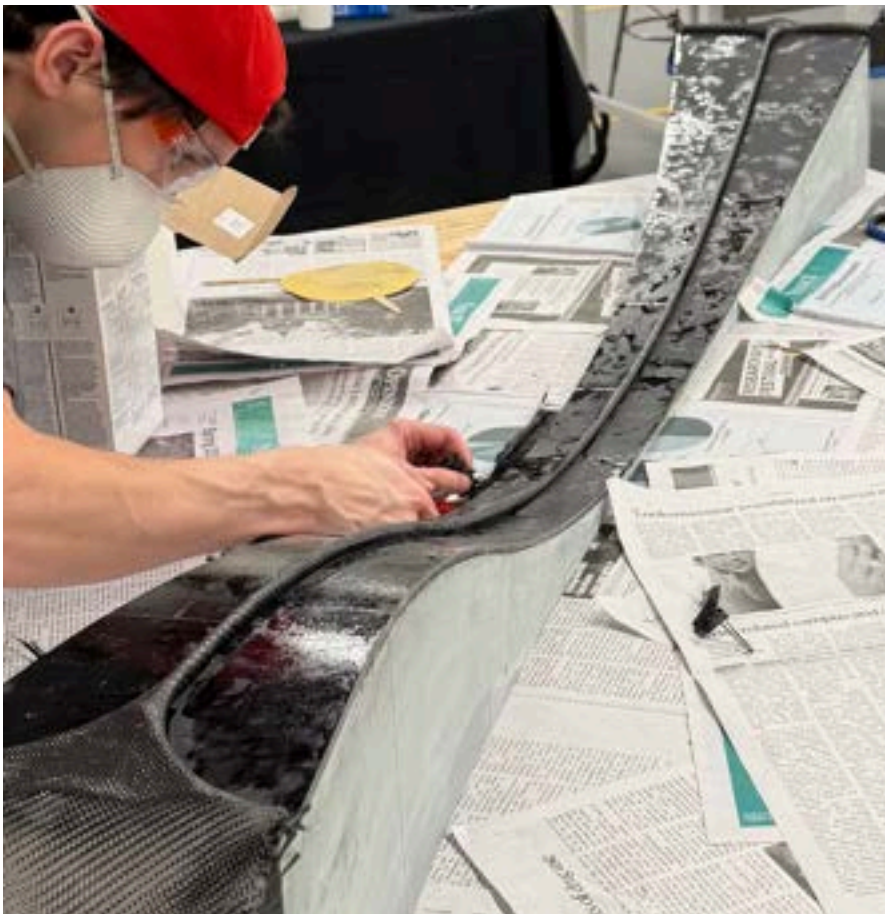
Strake mold, negative (gray) and positive (black):



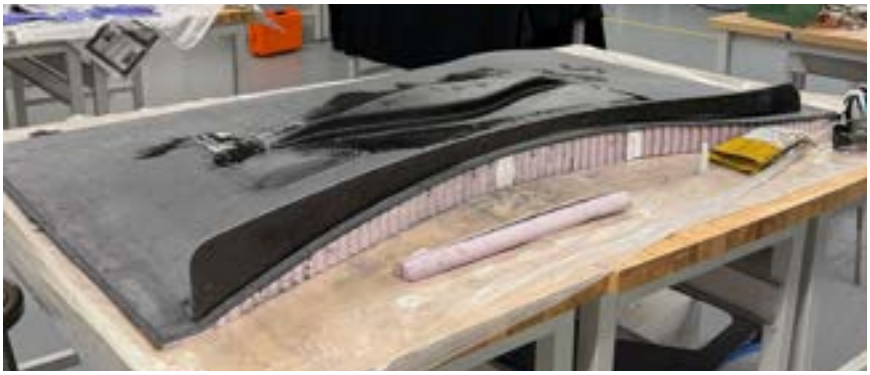
Preparing carbon for strake layups:



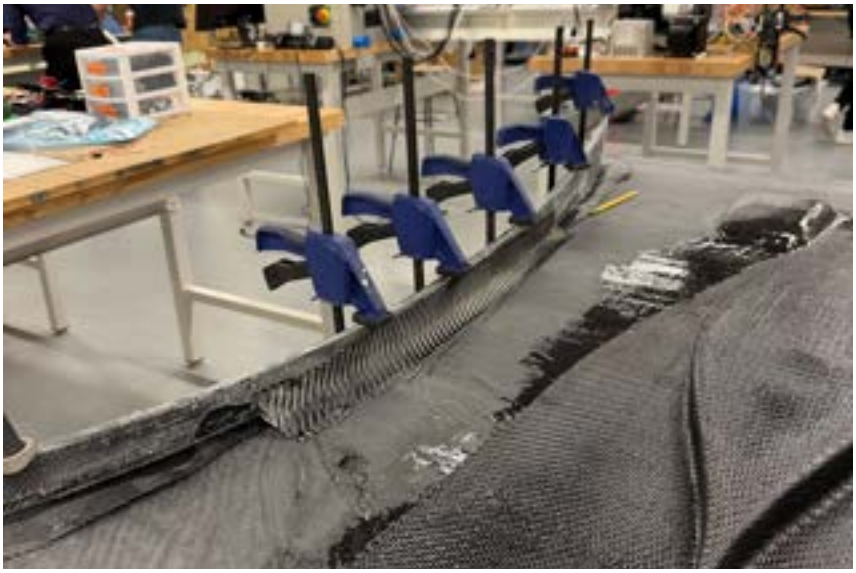
Strakes after initial layup, trimming the sharp edge down:



Strake being referenced to undertray after layup completed and separated from the mold:



Strake being attached to undertray, including extra fabric used to secure center:



Alternate angle of strake being attached to undertray:



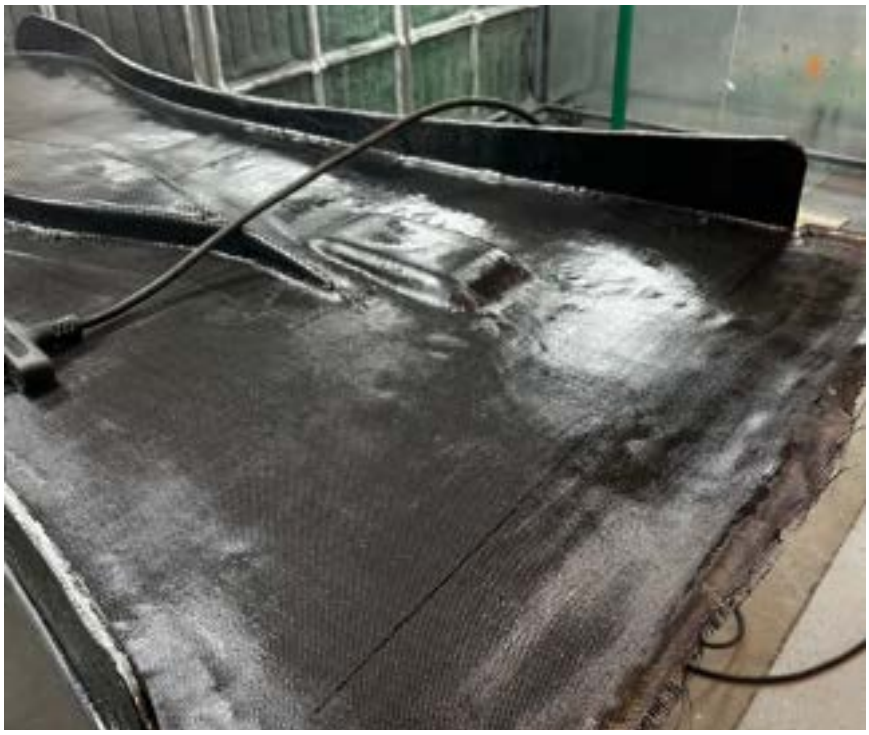
Second strake being attached to undertray:



Undertray after having the final layer of carbon applied to the underside:



Line marking excess carbon to be removed from undertray:



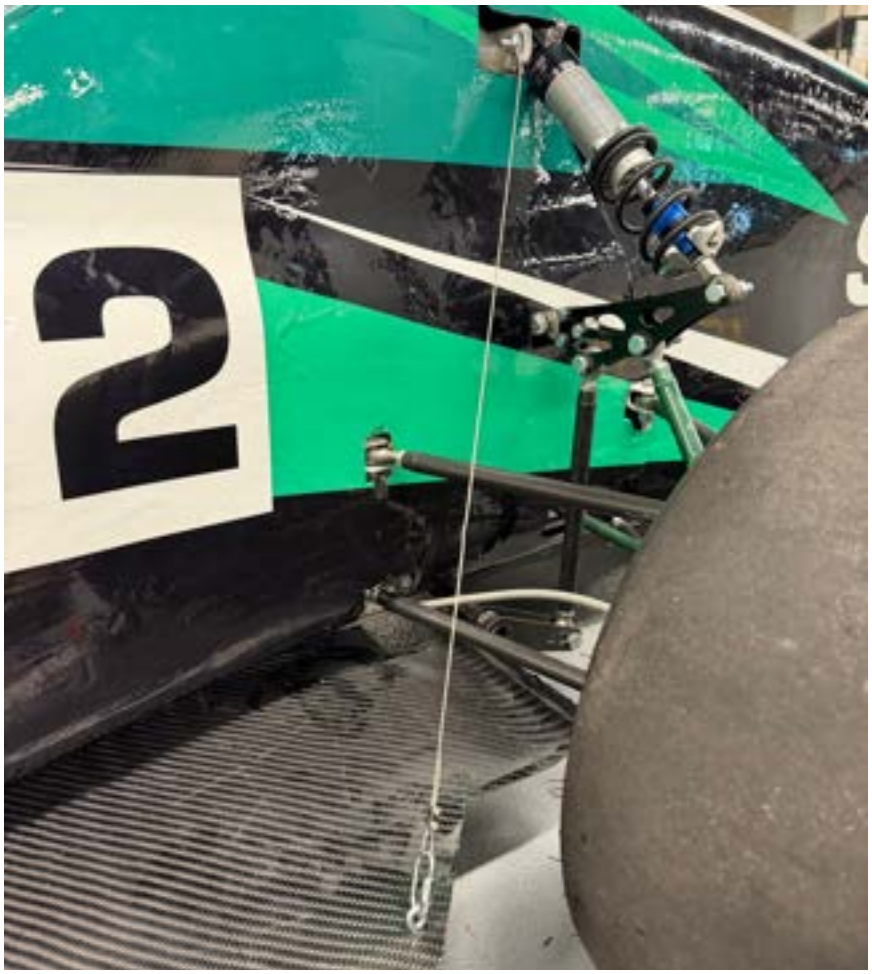
Minor excess material outboard of the strake before being trimmed off:



Undertray weight before being trimmed for front wheel turning clearance.



Undertray wire-hanger assembly, driver's right-hand side



Undertray mounted in final position on Binghamton Motorsports FSAE EV2

